Assignment-Dependent Charging and Marginal-Price Support in Electric-Bus Fleets

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Abstract

Service assignments change both an electric-bus fleet's charging opportunities and its vehicle cost. We derive an exact joint period-load–vehicle-cost projection for heterogeneous services in two batches with common charging windows, conversion efficiencies, a fixed successor graph and full replenishment. When finite repeated charge/discharge actions are allowed without switching or cycling penalties, one minimum-requirement matching value per cardinality determines a polygon with at most five vertices. A constructive serial realization respects individual inventories and the shared gross connector limit. Retaining these complete polygons preserves projected planning and price-response values even for negative prices; monotone supply costs admit a smaller efficient frontier with at most two segments per cardinality. The lossless specialization gives a scalar own-price support criterion. Matching, conversion accounting and price-support machinery are classical. Our physical reduction also has a precise limitation: edge-dependent compulsory transition consumption can retain a positive support gap on a complete assignment graph even when every perfect matching is feasible. The obstruction persists with strongly convex separable supply cost and in a relative open numerical neighborhood under fixed timing and access. Separately checked rational certificates establish a positive gap on one 37-service timetable under explicitly distinguished binary and exact-additive consumption models. These are model-specific existence results with synthetic coefficients, not evidence of operational prevalence or calibrated savings.

1 Introduction

An electric-bus operator assigns mandatory services to vehicles and decides when to recharge them. Continuous charging offers flexibility, but a vehicle cannot cover a fraction of a service. Assigning two services to the same bus can therefore impose an intermediate energy requirement that disappears when two buses are used. A tariff may reward the resulting change in charging without compensating the operator for its additional vehicle.

We ask a precise question. A supplier has a convex incremental cost for serving aggregate fleet demand. A tariff is calculated from a candidate fleet's load and posted before the operator chooses its complete schedule. Holding that tariff fixed, does the candidate minimize the operator's bill plus its intrinsic operating cost? This is an own-load price-support test. Its participant is one operator controlling the complete fleet. It is not a strategic model in which that operator anticipates a new tariff after each alternative it considers. The support test is classical; our contribution is the physical set description that makes its application explicit for a restricted fleet class.

The difficulty need not be a complicated load-only feasible set. Assigning services to the same bus can impose an early recharge requirement while reducing the number of paid vehicles. A fixed-population charging aggregate does not retain that tradeoff. We identify a restricted physical

class in which it can be described exactly: heterogeneous two-batch services, common charging windows and conversion efficiencies, a fixed successor graph, shared gross connector use and full terminal replenishment. A minimum-requirement matching at each cardinality suffices to construct the joint energy–vehicle-cost set. With losses and unrestricted finite early direction changes, this set is a union of polygons having at most five vertices each. Their constructive realization uses sequential actions, never simultaneous charging and exporting. Standard convexification then gives planning and own-price support decisions. For monotone objectives, removing dominated cycling actions leaves at most two efficient segments per cardinality. Keeping the full polygons avoids that economic restriction, but relies on ideal switching and period-aggregate settlement. Gross connector use counts both purchases and exports; full terminal replenishment prevents free net use of initial battery inventory.

We delineate why this reduction works. A common transition-energy task can be included because total energy stays fixed within a matching cardinality. Edge-dependent tasks break that property. A complete-graph construction admits every perfect matching yet retains a positive gap, even with strongly convex supply cost and strict feasibility margins. Its relative-neighborhood statement keeps the discrete topology, charging access and ideal common-window physics fixed. The example rules out several simple explanations for the obstruction without claiming generic routing hardness, exponential hull complexity or empirical robustness. V2G, paid idle fleets and ownership examples clarify further consequences; their complete arguments are retained in the relevant sections and appendices.

The public-timetable analysis serves a different purpose. Exact physical lower bounds and independently admitted complete-plan mixture uppers establish existence beyond the two-batch class. The original binary-consumption certificate concerns its declared movement graph. A separately defined exact-additive law, fresh target certificates and a dominance argument cover finite walks over specified paired arcs and continuous waiting. These distinct models share synthetic economics and do not establish empirical road completeness or operational prevalence. Small exact qualification suites and the original-model portable verifier address implementation trust, not generalization or third-party replication.

2 Relation to existing work

Operational coupling. Grid-aware scheduling, capacitated charging and scalable route-and-recharge models already combine bus duties with charging access [22, 9, 17]; integrated flow and set-partitioning models also choose assignments and charging slots [7, Sections 2.1 and 4]. Wu et al. [22] include endogenous vehicle-to-trip assignment and full overnight charging. Integer bus-charging aggregation/disaggregation around auction bids is studied by Zoltowska and Lin [25], while Wu et al. [23] couple service planning to market clearing and distribution marginal prices in a bilevel model.[1] Thus assignment, full charging and endogenous electricity prices are not sufficient novelty claims. We instead ask whether a fixed posted marginal tariff supports a complete fleet over alternatives that change both used-bus costs and duty-created charging opportunities.

Transport and V2G are also established: Blatiak et al. [6, Sections 2.1–2.2] optimize commercial-fleet trip timing and charging against energy and balancing prices. The author manuscript of Zhou et al. [24, Section 2] chooses bus use, connections and partial charging/discharging under input purchase/export tariffs, including vehicle fees, deadheading, charging, export revenue and degradation, with full-depot inventory/overnight recharging. These materially overlap our physics;

[1] Our comparison with Wu et al. [23] is limited to the publisher's preview. Its full tariff timing and complete-fleet deviation formulation remain unverified; we do not infer their absence.

the manuscript's imperfectly extracted mathematical glyphs preclude detailed claims about its connector or terminal constraints. Our distinction is an exact cost-labelled threshold and own-load support classification, not richer V2G operations, first transport–V2G coupling or first use of spare buses for grid services.

Joint loads and vehicle cost. Mukhi and Abate [14] characterize fixed-population charging polytopes with prescribed availability and full departure charge using permutahedra. Mukhi et al. [15] give generalized-polymatroid representations and exact Minkowski aggregation for convex individual flexibility sets. Arvis et al. [4] integrate prescribed EV flexibility, including V2G, into unit commitment using exact aggregation and separation. Coupling vehicle flexibility with grid dispatch is therefore not itself our novelty. Al Taha et al. [2, Section II, Example 1] likewise aggregate lossless bidirectional batteries with initial/final requirements. Our lossless load-only projection is an interval, not a new nonconvex flexibility set. Theorem 5 instead retains every attainable load–vehicle-cost layer as successor matching and fleet size change; Corollary 6 tests hull-optimum physical attainment. Fixed-matching interval aggregation and ordered matching remain classical. The V2G thresholds and capped paid-idle elimination concern this restricted class with shared gross throughput; persistence/restoration examples do not establish a first signed-aggregation or V2G support result.

Common losses. Al Taha and Bitar [1, Theorems 1–3] establish a piecewise-affine grid-power/SOC bijection, convex feasibility in a single store's SOC variables, and sufficient monotonicity conditions for convex reformulation. These transformations and conditions are prior art. Their discrete signed control has one direction per time step. Our complete projection explicitly permits successive actions inside one common window; its purchases and exports are integrated totals, not simultaneous physical controls. Dropping complementarity alone would not establish exactness. Theorem 12 instead constructs serial actions respecting SOC and joint polygons for loads and vehicle cost, with at most five vertices per matching cardinality. Theorem 16 gives the smaller efficient frontier with at most two segments. Common windows and monotone economics are essential to discarding dominated points in that smaller representation. Conversion bookkeeping, time sharing, box aggregation and planar vertex counting are classical; the physical replacement across all assignments is the claim here.

Support, lost opportunity cost and ownership. Nonconvex pricing and uplift are established [16, 10]; convex-hull pricing supplies the convexification and lost-opportunity-cost framework [18, 12]. Complete linear descriptions address support with integralities [13], and whole-participant schedule columns with fixed-price self-scheduling oracles have established decomposition treatments [3]. Our support equivalence and complete-fleet unit apply these tools, not new pricing duality or column-generation machinery. Classical aggregation [21, 11] explains small aggregate nonconvexities; our replication calculation distinguishes the planning gap, a composite owner's absolute fixed-price incentive and independently reserved pairs' incentives. Its rates do not contradict approximate aggregation results.

The contribution is a physical classification under explicit assumptions, not a richer operational model or faster general scheduling algorithm. Matching integrality, value-function convexity and price-support duality remain the underlying tools. Capacity, fee and equipment examples explain the classification; the public certificate establishes existence within its declared movement graph, without extending the closed-form theorem to arbitrary timetables.

3 Complete fleets and the meaning of support

Fix mandatory services, directed movement opportunities, a charging horizon and physical equipment. A complete plan $x \in \mathcal{X}$ covers every mandatory service exactly once, assigns the services to used buses, routes those buses, and schedules charging subject to battery and shared-resource constraints. Uncounted buses neither exchange energy nor incur operating cost. When a variant admits paid idle buses, they count as used participants under that variant's fleet and access rules. Let $e(x) \in \mathbb{R}^T$ denote net grid energy by market interval, positive for purchases and negative for exports, and let $c(x)$ be intrinsic operating cost. Charge-only variants restrict $e(x) \in \mathbb{R}^T_+$. Each used battery starts full and finishes at the same required full inventory. Assume the projected image

$$S = \{(e(x), c(x)) : x \in \mathcal{X}\}$$

is nonempty and compact. All physical planning, price-response and hull problems below use exactly this common set.

Let $F : \mathbb{R}^T \to \mathbb{R} \cup \{+\infty\}$ be proper, closed and convex, finite on all feasible loads. It represents incremental supply cost conditional on fixed background demand, on the declared feasible signed-load domain when exports are allowed. Define

$$D = \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \tag{1}$$

$$\mathrm{CH} = \min_{(e,c) \in \operatorname{conv} S} \{c + F(e)\}, \qquad \Delta = D - \mathrm{CH} \geq 0. \tag{2}$$

The hull mixes *complete feasible plans*; it is not a duty-column relaxation unless equivalence is separately proved. Its mean load need not be executable as a physical daily schedule. Evaluating supply cost at that mean also differs from averaging the costs of plans operated on different days.

For a posted tariff $p \in \mathbb{R}^T$, the complete-fleet response and regret are

$$V(p) = \min_{y \in \mathcal{X}} \{c(y) + p^\top e(y)\}, \qquad r(x; p) = c(x) + p^\top e(x) - V(p). \tag{3}$$

The posted tariff is fixed across alternatives y. A strategic operator anticipating a tariff reset would face a different objective and needs a separate settlement model.

Definition 1 (Weak own-load support)**.** A physical plan x is supported if there exists an admissible marginal price $p \in \partial F(e(x))$ for which $r(x; p) = 0$. Alternative plans may tie. If an economic convention restricts the price set, existence is tested only in that declared subset.

For differentiable supply we use its gradient. For our quadratic examples this means the gradient of the unrestricted quadratic formula, including at a zero load. In charge-only variants, orthant-domain normals are not additional generation-price choices. With nonsmooth supply, a failed support test at one chosen multiplier does not exclude support at another admissible price.

For the following classical equivalence, assume either

$$\operatorname{ri} \operatorname{conv} e(\mathcal{X}) \cap \operatorname{ri} \operatorname{dom} F \neq \varnothing,$$

or that a finite convex minorant on $\mathbb{R}^T$ agrees with F on all hull loads. The latter applies to our quadratic. These assumptions ensure an attained hull-dual price; a restricted economic price set must contain such a price for the corresponding converse.

Proposition 2 (Classical support criterion)**.** *For every physical x and $p \in \partial F(e(x))$,*

$$r(x;p) \geq c(x) + F(e(x)) - \mathrm{CH} \geq \Delta. \tag{4}$$

Under the stated qualification, $D = \mathrm{CH}$ if and only if some physical plan is supported. If $D = \mathrm{CH}$, every physical planner optimum is supported by every admissible *attained maximizer of $V(p) - F^*(p)$, where $F^*(p) = \sup_u \{p^\top u - F(u)\}$.*

This specializes established nonconvex support and convexification arguments [16, 10, 12]; the proof is included in Appendix O.1 to make assumptions and quantifiers explicit. It does not say that every subgradient supports a planner optimum, or that a price-adjustment algorithm converges. Without a price restriction, every attained dual maximizer has zero regret and is a subgradient at each physical planner optimum. An external price restriction adds a separate admissibility test to that mathematical conclusion.

Remark 3 (Admissibility is a separate quantifier)*.* For a single physical point $(e, c) = (0, 0)$ and $F(e) = |e|$, both values are zero and every $p \in [-1, 1]$ attains the hull dual. If the declared admissible price set is $\{0\}$, the plan has admissible support at zero; the dual maximizer $p = 1$ remains mathematically supporting but is inadmissible. Containing one dual maximizer does not mean containing them all.

The participant in (3) controls all buses. Shared infrastructure does not automatically define unilateral alternatives for separately owned vehicles. Section K introduces reserved rights before making an individual-participant comparison.

4 An exact assignment-dependent physical projection

The central result retains vehicle cost together with grid load. It applies to an ideal common-window class, rather than arbitrary operational timetables. There are $n \geq 1$ mandatory A services in $[0, 1]$ and n mandatory B services in $[2, 3]$, with battery demands $0 < a_i, b_j \leq B$ and common capacity $B > 0$. A fixed bipartite graph Γ lists every allowed A–B successor pair. Every service also has a standalone option; transitions have zero time, energy and monetary cost. A partial matching of size m uses $N_m = 2n - m$ buses, each at fee $f > 0$, under fleet cap $2n$. Services are indivisible noncharging intervals with nonnegative battery draw throughout.

All buses start full, finish full and are available throughout the early window $[1, 2]$. One shared connector has grid-side power $K \geq 0$ in that window, in either direction; terminal charging uses one connector of power $H \geq 0$ in $[3, 4]$ and permits charging only. Every bus accepts the corresponding powers. Charging efficiency is common $\alpha \in (0, 1]$ in both windows, and export efficiency is common $\delta \in (0, 1]$. Reserve and other auxiliary draw are zero. Early actions may change direction finitely many times, with no switching delay, minimum session, degradation or cycling penalty. They are sequential, never simultaneous on the connector. These permissive action assumptions matter for exactness.

Let C, X be integrated early grid purchases and exports. The connector counts gross use $C + X \leq K$, while net early load is $E = C - X$. Write

$$S_0 = \sum_i a_i + \sum_j b_j, \quad P = \sum_i a_i, \quad Q = \alpha C - X/\delta.$$

Here Q is net battery exchange, and full replenishment requires terminal grid purchase $L = (S_0 - Q)/\alpha$. Appendix derivations use the local symbol S for this service-energy total; S_0 distinguishes it from the framework's projected set. Supply and tariff objectives must be defined on the resulting signed load domain, or a declared background translation.

Theorem 4 (Complete common-window projection)**.** *Let ν be the maximum matching size in Γ, and define*

$$r_m = \min_{\substack{M\subseteq\Gamma \\ mmatching \\ |M|=m}} \sum_{(i,j)\in M} (a_i + b_j - B)_+, \qquad r_0 = 0, \qquad m = 0, \dots, \nu.$$

For each m, let

$$W_m = \{(C, X) : C \geq r_m/\alpha, \quad X \geq 0, \quad C + X \leq K, \\ \max\{S_0 - N_m B, S_0 - \alpha H\} \leq \alpha C - X/\delta \leq P\}.$$

The complete physical load–vehicle-cost projection equals

$$\mathcal{P} = \bigcup_{m=0}^{\nu} \left\{ \left(C - X, \frac{S_0 - Q}{\alpha}, fN_m \right) : (C, X) \in W_m \right\}.$$

Each point has a finite-action physical realization. Empty polygons are discarded; each remaining polygon has at most five vertices. Their affine images generate $\operatorname{conv}\mathcal{P}$ *with at most* $5(\nu+1)$ *records, including degenerate and lossless images.*

Why one matching statistic suffices. For a fixed matching, each bus's net battery increment has an interval: paired buses use $[a_i+b_j-B, a_i]$, A-only buses $[a_i-B, a_i]$, and B-only buses $[b_j-B, 0]$. Their lower endpoints sum to $S_0 - N_m B$, upper endpoints to P, and positive lower endpoints to the matching's requirement R. At any admissible Q, interval allocation realizes positive increments totalling $A_+ = \max\{R, Q\}$. One-direction actions therefore use $C_0 = A_+/\alpha$ and $X_0 = \delta(A_+ - Q)$.

Any proposed polygon point has residual purchases $z = C - C_0 \geq 0$ and residual exports $\alpha\delta z$. One existing bus can realize them by battery-neutral cycles before its baseline exchange. Since $N_m \geq n \geq 1$ and $B > 0$, choose charge-first or export-first according to its SOC; chunks of battery depth at most $B/2$ fit and return to that SOC. Finitely many chunks realize the finite residual. Serialize all actions: total early throughput fits K, and terminal refill fits H. Zero power permits only zero energy, without division. Positive B-service demands give $P < S_0$, so terminal purchases are nonnegative. At fixed m, only R depends on the matching; a matching attaining r_m contains every other matching's entire polygon at the same C, X and fee. This is physical equality, not just economic dominance. The complete proof is Theorem 12.

5 Decisions, boundaries and independently certified existence

The exact set preserves arbitrary objectives on period loads and cost, including negative posted-price response values. For convex supply F, physical planning minimizes $c + F(E, L)$ over each polygon and takes the smallest value; hull planning uses ordinary convex combinations of their vertex records. This uses classical optimization, matching and support theory, not a new convexification algorithm. Under the framework's nonemptiness assumption (at least one W_m is nonempty) and qualification, $D = \mathrm{CH}$ characterizes physical own-load support; a strict gap excludes it. Replacing complete-plan weights by a fractional fleet cardinality and interpolated r_m need not give the same hull.

Monotone supply objectives permit a smaller efficient description with at most two segments per cardinality (Theorem 16). It cannot replace the full polygons for arbitrary negative-price queries. The charge-only lossless scalar classification is given in Corollary 6. Common unequal

directional powers and affine grid-throughput wear also admit exact polygons, using weighted connector occupancy and retaining wear in the joint cost image (Corollary 14). Its positive-power finite action bound does not transfer the older efficient formulas or cover state-dependent degradation and switching time. Positive per-action setup time admits a conservative physical inner polygon and a quantitative value/gap bound under explicit connector-time slack (Proposition 33). Without slack, an arbitrarily small setup can remove a fleet-cost layer while other fleets stay feasible. Even when two perfect matchings remain feasible, setup can invalidate minimum-requirement matching replacement: the number of buses needing an early charge matters separately (Proposition 34).

A common compulsory post-early transition task preserves the reduction with $S_0 + md$ and a restricted graph $b_j + d \le B$ (Corollary 19). Edge-dependent tasks can break this sufficient statistic. A complete-graph construction admits every perfect matching yet retains a positive support gap, including with strongly convex supply and a relative open parameter neighborhood (Proposition 38). That neighborhood fixes topology, timing, access and ideal physics; it supplies no uniform radius or empirical robustness claim. The detailed boundary examples, lossless/V2G classifications and ownership consequences remain in the appendices.

One public timetable, two explicitly distinct energy laws. A separate 37-service case goes beyond the common-window class. Under each of two separately certified models, the conservative outward bounds are

$$\mathrm{CH} \le 441.142238 < 441.532912 \le D \le 441.532913, \qquad D - \mathrm{CH} \ge 0.390674.$$

The physical optimum is tightly enclosed, not known exactly; the hull optimum is also unknown. A feasible complete-plan mixture supplies its upper, while an all-physical rational branching certificate supplies the lower. A mean plan is not an executable daily schedule.

The original binary-consumption result concerns its finite declared movement graph. A separately defined exactly additive consumption law, fresh target certificates and a canonical dominance proof extend the second result to finite walks over specified paired OD arcs and continuous linear waiting, under explicit triangle and common-access premises. No certificate or feasibility conclusion transfers automatically between these laws, even though the displayed decimal bounds coincide. These are charge-only models with synthetic consumption/economics, 400 kWh batteries, one 360 kW connector and used-bus fee 100. They establish neither public V2G nor calibrated savings, prevalence or empirical road completeness. Section E and Proposition 20 give the full model, exact certificate scopes and extension conditions.

6 Implications and limits

The obstruction concerns one posted-tariff institution with a complete-fleet participant. It is a possibility under characterized conditions, not an inevitable failure of marginal tariffs, a strategic tariff-reset equilibrium or a convergence claim. Convex-hull pricing and general support duality are established theory. The physical contribution is the explicit assignment-dependent projection under the stated timing, resource, loss and action assumptions.

The exact analytical projections omit stochastic delays, state-dependent degradation, temperature, taper and switching time. A separate setup-time approximation bounds the effect of a positive per-action delay when every relevant layer has verified connector-time slack; without that premise, even value continuity can fail (Proposition 33). A common affine-throughput wear extension remains an ideal cost model, not a battery-aging calibration. Common windows and efficiencies

enable the finite reduction; arbitrary timetables, vehicle-dependent losses, restricted cycling and intrawindow price variation require further analysis. Full polygons can handle negative period prices; the smaller efficient frontier cannot discard larger loads under a nonmonotone objective. The nominal reserve/loss neighborhood does not validate those abstractions empirically. The public case uses varied times and movement energy, but remains charge-only within fixed declared movement modes and charging windows. Its certificate does not prove completeness for arbitrary retiming or unlisted access. A separately defined exact additive law admits a wider result: under its explicit triangle and charging-access premises, a separate certificate covers arbitrary finite walks in the paired-arc model. Neither law certifies all empirical road options. Synthetic coefficients and one base timetable leave calibrated magnitude and prevalence open.

7 Conclusion

Service assignments change the joint load–vehicle-cost set even when its load-only projection is simple. For the stated common-window class, cardinality-matching values determine the complete load–cost projection with common losses and ideal finite cycling. The lossless specialization gives an exact scalar support classification; monotone objectives admit a smaller efficient frontier. Gross connector use, full replacement energy and constructive serial attainment are essential to that description. A separately checked public-timetable certificate establishes the same qualitative obstruction in a richer declared charge-only model. These results identify when continuous charging flexibility does and does not remove assignment-induced price-support failure, while leaving empirical significance and more general scheduling classes open.

Evidence and reproducibility

Independent physical enumeration qualifies the finite analytical implementations; constructive proofs establish their continuum statements. The full common-loss polygons separately matched raw bus-level enumeration on eight fixed cases, including finite-cycle replay and common transition energy; earlier one-direction checks have a narrower scope. Appendix Y preserves all case/control denominators and their scopes. The public result uses classical native proposals followed by exact physical admission and rational coverage certificates. One fresh standalone standard-library Python process rechecked the original binary-model proof without a solver in a construction-and-check phase of about 33 seconds. That exported checker does not certify the separate exact-additive model. This is internal author-run verification, not external replication. The certificate index binds sources, inputs, exact constants and command. Earlier failures and costs remain archived. The exported package stays local pending redistribution review; no public release or substantive learning contribution is claimed.

A Complete assignments and their load–cost frontier

The relevant projection keeps operating cost as well as load. A whole interval of early loads can be physical while the cheapest vehicle count changes discontinuously across it. We characterize this joint frontier under fixed shared resources and the declared allowed successor graph. A separate adequate-resource policy is retained in Appendix H.

There are $n \geq 1$ mandatory depot-to-depot services A_i in $[0,1]$ and n services B_j in $[2,3]$. Batteries have capacity $B > 0$ and $0 < a_i, b_j \leq B$. A fixed declared bipartite graph Γ specifies which A-to-B shared-bus assignments are allowed. Every allowed transition has zero movement time, energy and cost. Every service retains its standalone bus option and its original depot charging access. At most $2n$ buses are used, each at fee $f > 0$, starting full and returning full by hour 4. One shared early connector has power $K \geq 0$ in $[1,2]$; one shared terminal connector has power $H \geq 0$ in $[3,4]$. Each bus accepts the corresponding connector power. There is no other charging, reserve, loss, taper, auxiliary draw or switching time. Powers K, H are fixed resources, not multiples of n. The graph is an operational compatibility restriction, not a selected bank of duties used to approximate a larger declared opportunity set. Write ν for its maximum matching cardinality, including $\nu = 0$.

Define

$$S = \sum_i a_i + \sum_j b_j, \quad L = \max\{0, S - H\}, \quad U = \min\{\sum_i a_i, K\}, \quad r_{ij} = [a_i + b_j - B]_+,$$

$$r_m = \min_{\substack{M \subseteq \Gamma \\ |M| = m}} \sum_{(i,j) \in M} r_{ij}, \qquad r_0 = 0, \qquad c_m = f(2n - m),$$

where M ranges over allowed partial bipartite matchings and $m = 0, \ldots, \nu$. Every such cardinality exists by deleting edges from a maximum matching. A matching of size m uses $2n - m$ buses.

Theorem 5 (Fixed shared-resource load–cost projection)**.** *The model is feasible exactly when $L \leq U$. In that case put*

$$m_{\max} = \max\{0 \leq m \leq \nu : r_m \leq U\}, \qquad m_0 = \max\{0 \leq m \leq \nu : r_m \leq L\}.$$

Its complete physical projection is

$$\bigcup_{m=0}^{m_{\max}} \{(E, S - E, c_m) : \max\{L, r_m\} \leq E \leq U\}. \tag{5}$$

The thresholds r_m are nondecreasing and discretely convex. The lowest operating-cost boundary $C(E)$ of the complete-plan hull starts at (L, c_{m_0}), visits (r_m, c_m) for $m_0 < m \leq m_{\max}$, and is horizontal at $c_{m_{\max}}$ on $[\max\{L, r_{m_{\max}}\}, U]$.

For a fixed matching, early purchases satisfy $0 \leq x_i \leq a_i$ and $x_i \geq r_{ij}$ for a paired service. Their sum ranges over $[R(M), \sum_i a_i]$. Serialize them at power K: the shared early constraint is exactly $E \leq K$. Every bus is present for terminal charging, and individual refill demands sum to $S - E$. Serialization at power H is possible exactly when $S - E \leq H$. Zero-power cases require zero energy and no division. Appendix L proves both directions and the frontier geometry. Cardinalities above ν do not exist; those with $r_m > U$ are resource-infeasible. Neither supplies a fictitious upper-clipped endpoint. The energy-only feasible set is simply $[L, U]$; it is the joint cost frontier that matters for convexification.

Corollary 6 (Support from the cost-labelled frontier)**.** *Suppose $L \le U$ and $G(E) = F(E, S-E)$ is finite, continuous and strictly convex on $[L, U]$. Let E^* uniquely minimize $C(E) + G(E)$. Then*

$$D = \mathrm{CH} \quad \Longleftrightarrow \quad E^* \in \{L\} \cup \{r_m : m_0 < m \le m_{\max}\} \cup [\max\{L, r_{m_{\max}}\}, U]. \tag{6}$$

The gap is positive precisely in the open intervals between adjacent retained thresholds. Admissible marginal-price support additionally requires Proposition 2's supply and price-attainment qualification.

The clipped endpoint L and intermediate physical thresholds on collinear edges must be retained. If $L = U$ or $m_0 = m_{\max}$ the gap is zero. At $f = 0$ the zero-matching branch attains every load at zero operating cost, so the positive-fee interior criterion would fail. The empty matching remains available even with isolated vertices or no allowed edges. Consequently the feasibility test is unchanged; when $\nu = 0$, the entire projection has operating cost $2nf$ and its gap is zero.

Corollary 7 (Classical ordered matching reduction for the complete graph)**.** *If Γ contains every A-to-B edge, then for increasingly sorted energies $a_{(i)}, b_{(i)}$,*

$$r_m = \sum_{i=1}^{m} [a_{(i)} + b_{(m+1-i)} - B]_+.$$

Monotonicity permits selecting the m smallest energies in each batch; a convex positive-part exchange inequality then favors opposite pairing. The appendix proves this classical assignment reduction. No matching algorithm novelty is claimed. The shortcut does not extend to arbitrary allowed graphs. With A energies $(5, 15)$, B energies $(14, 15)$ and battery capacity 20, allowing only the two corresponding-index edges gives $r_2 = 10$; the complete graph's crossed assignment gives $r_2 = 9$. The general theorem uses the graph's actual minimum-cost matching values and never invents a forbidden edge.

The fixed allowed-graph extension has independent source-only proof review; its sparse-graph implementation qualification has not been performed. The complete-crossmatching specialization has a separate, prospectively frozen fourteen-case exact implementation qualification. Raw SOC enumeration and serial charging witnesses matched the sorted-threshold formulas in all fourteen cases; all twenty-four negative controls and the smoke anchor passed. The cases include a physically attained intermediate collinear threshold and a supported clipped lower boundary, as well as zero-power, zero-fee, infeasible and continuous-response cases. This is an internal implementation check, separate from the eight-case adequate-resource qualification in Appendix H. Fixed chargers also impose throughput limits as service demand grows; this is no claim of indefinite scalability with n. Nonzero transition consumption would make total energy matching-dependent; transition time or staggered charging windows would invalidate the common- window serial argument. Inserting an arbitrary directed timetable graph without addressing those changes is therefore outside this theorem. Even a third batch of simultaneous services can change the combinatorics: with no intermediate charging, complete transition compatibility and ample terminal charging, fleet feasibility is already strongly NP-complete by a direct numerical three-dimensional-matching reduction (Appendix M). This classical boundary explains why pairwise matching need not characterize longer duties.

A.1 A fixed single-connector heterogeneous obstruction

Take $n = 2$, $B = 20$, $f = 7$, A energies $(5, 15)$ and B energies $(14, 15)$, with every crossmatching admitted. Use one early connector of power $K = 15$ and one terminal connector of power $H = 49$.

The early-requirement matrix is

$$(r_{ij}) = \begin{pmatrix} 0 & 0 \\ 9 & 10 \end{pmatrix}.$$

The crossed full matching requires 9 early kWh; the corresponding-index full matching requires 10. Thus $r_0 = r_1 = 0$, $r_2 = 9$, $S = 49$, $L = 0$ and $U = 15$. Theorem 5 gives the complete physical projection:

Matched services m	Early energy interval	Operating cost
0	$[0, 15]$	28
1	$[0, 15]$	21
2	$[9, 15]$	14

Use the convex supply

$$F(E, L) = 4E + \frac{E^2 + L^2}{20}, \qquad G(E) = F(E, 49 - E) = \frac{2401}{20} - \frac{9E}{10} + \frac{E^2}{10}.$$

The physical optimum is the crossed two-bus plan at $E = 9$. The hull joins $(0, 21)$ to $(9, 14)$ and minimizes at $E^* = 151/18$, strictly inside that edge. Exactly,

$$D = \frac{2681}{20}, \qquad \mathrm{CH} = \frac{434201}{3240}, \qquad \Delta = \frac{121}{3240} > 0. \tag{7}$$

At the physical optimum's marginal price $(49/10, 4)$, its bill plus operating cost is $2181/10$; the one-pair zero-early complete fleet costs 217, so its full-opportunity-set regret is $11/10$.

Both hull endpoints have serial full-inventory witnesses under these fixed shared resources. At zero early charge, pair the 5-energy A with the 14-energy B and serve the other two services separately. Its three terminal demands 19, 15 and 15 total 49 and serialize on the one 49 kW connector in the terminal hour. At early charge 9, use the crossed full matching: the 15-energy A bus charges for 9/15 hour on the one 15 kW early connector. Both buses then require 20 terminal kWh, serializing in 40/49 hour on the terminal connector. Every used bus starts and ends full. Weights $11/162$ and $151/162$ on these two complete plans give the hull point. The mean is a whole-plan convexification, not a physical fractional fleet.

The fixed shared-resource, complete-crossmatching implementation was separately qualified on fourteen prospectively fixed exact cases, including this smoke anchor, collinear thresholds, clipped boundaries, zero powers, fee zero, infeasibility and full response continua. All twenty-four negative controls passed. This is internal qualification, not operational prevalence; it does not qualify sparse allowed graphs or the later V2G extension. Appendix H preserves the different n-parallel- connector policy, its continuous value formulas and its separate eight-case qualification. It has the same joint projection for this example but different individual allocation sets; none of its parallel witness timings is used here.

The main theorem admits fixed shared powers and an arbitrary declared allowed graph, but retains two simultaneous batches and zero-time, zero-energy allowed transitions. It does not cover the public varied-time case. The following two-service model makes the same physical/hull distinction transparent and shows how connector and equipment changes can preserve or remove it.

B Bidirectional early charging under an explicit settlement rule

The two-batch model also permits an exact extension in which assigned buses can discharge before the second batch. This changes both the physical opportunity set and the tariff interpretation. We

assume ideal efficiency one, symmetric charging and discharging acceptance, and the *same* posted price for signed net purchases and exports in a market interval. An asymmetric buy/sell tariff or a lossy battery is a different model.

Retain n A services in $[0,1]$, n B services in $[2,3]$, capacity $B>0$, consumptions $0<a_i,b_j\le B$, and the fixed allowed successor graph Γ of Section A. Every singleton remains available; allowed transitions have zero time, energy and cost. A matching of size m uses exactly $N_m=2n-m$ assigned buses, each with fee $f>0$, full initial inventory and full terminal replenishment. We first derive the projection with service-assigned buses, then allow identical paid idle buses below. In particular, a B-only bus is assigned to service, starts full and is available at the depot throughout the early window.

All assigned buses can use the single early connector during $[1,2]$ in either direction at power magnitude $K\ge 0$. Its unit-window resource is *gross* throughput: if bus v has signed net early purchase q_v, then $\sum_v |q_v|\le K$ must be realizable. A constraint on $|\sum_v q_v|$ alone would incorrectly allow charging and discharging to cancel connector time. Terminal replenishment in $[3,4]$ uses one charging-only connector of power $H\ge 0$. Each bus accepts those corresponding powers. There is no other charging, reserve, auxiliary draw, taper or switching time.

Net early load $E=\sum_v q_v$ may be negative. Consistent with the signed framework in Section 3, F must be defined and finite on all admitted signed loads $(E,S-E)$. Alternatively, declare fixed background demand and an incremental cost $F(e)=C(d+e)-C(d)$ whose domain contains the whole feasible signed-load set. Export settlement still uses the same linear rate as purchases. There is no implicit extension of a cost defined only on nonnegative fleet demand. If aggregate export is prohibited, impose $E\ge 0$ explicitly; the lower thresholds below are then additionally clipped at zero.

Write $S=\sum_i a_i+\sum_j b_j$, $P=\sum_i a_i$, and let ν be the maximum matching cardinality of Γ. Retain the matching values

$$r_m=\min_{\substack{M\subseteq\Gamma\\ |M|=m}}\sum_{(i,j)\in M}[a_i+b_j-B]_+,\qquad 0\le m\le\nu,$$

with $r_0=0$. The matching machinery is unchanged; the available assigned battery inventory and gross connector budget create the new thresholds.

Lemma 8 (Minimum gross energy for a fixed matching)**.** *For a fixed allowed matching M of size m, put*

$$R=\sum_{(i,j)\in M}[a_i+b_j-B]_+,\qquad T=S-N_mB.$$

Before the shared connector constraint, signed early charges have total $E\in[T,P]$. At a prescribed such E, their minimum gross throughput is

$$\min_{\sum_v q_v=E}\sum_v|q_v|=\max\{E,2R-E\}. \tag{8}$$

The minimum is attained by one monotone action of the chosen sign per bus.

Theorem 9 (Signed-load projection and joint frontier)**.** *Under the above assumptions, define*

$$t_m=\max\{S-H,\ S-(2n-m)B,\ 2r_m-K\},\qquad U=\min\{P,K\},\qquad c_m=f(2n-m). \tag{9}$$

The complete physical projection is exactly the union of

$$\{(E,S-E,c_m):t_m\le E\le U\} \tag{10}$$

over $0 \le m \le \nu$ with $t_m \le U$. The model is feasible exactly when $t_0 \le U$. The sequence t_m is nondecreasing and discretely convex.

When feasible, let $m_{\max} = \max\{m \le \nu : t_m \le U\}$ and $m_0 = \max\{m \le m_{\max} : t_m = t_0\}$. The hull's lowest operating-cost boundary starts at (t_0, c_{m_0}), visits (t_m, c_m) for $m_0 < m \le m_{\max}$, and is horizontal at $c_{m_{\max}}$ on $[t_{m_{\max}}, U]$. Physical thresholds on collinear hull edges remain part of this statement.

Corollary 10 (Support and physical attainment)**.** *Suppose $t_0 \le U$ and $G(E) = F(E, S - E)$ is finite, continuous and strictly convex on $[t_0, U]$. Let E^* be the unique minimizer of the hull operating boundary plus G. Then, for $f > 0$,*

$$D = \mathrm{CH} \quad \Longleftrightarrow \quad E^* \in \{t_0\} \cup \{t_m : m_0 < m \le m_{\max}\} \cup [t_{m_{\max}}, U].$$

Admissible own-load support additionally requires the framework's convex- supply, dual-attainment and price-set qualification on the declared signed domain. At $f = 0$, the empty-matching branch has zero operating cost over the entire feasible interval and the gap is zero, so the positive-fee strict-interior criterion must not be applied.

Appendix N gives the constructive inventory, serialization and frontier proofs, including zero-power and zero-fee boundaries.

B.1 Identical paid idle buses

An idle bus has the same capacity, initial full inventory, terminal-full requirement, connector access and fee as a service bus. Its early signed charge lies in $[-B, 0]$ and its terminal refill is $-q$. It has no separate energy source or export tariff.

Proposition 11 (Idle buses under a total fleet cap)**.** *Allow identical paid idle buses, with total fleet at most $2n$. The complete cost-labelled projection in Theorem 9 is unchanged.*

The edge-deletion proof in Appendix N uses the total-fleet cap and identical battery, fee and access assumptions.

For the two examples below, $H = S$. Even if the fleet cap is removed, fleets larger than $2n$ cannot improve the lower frontier: terminal capacity forces $E \ge 0$, and the $2n$ standalone buses realize every $E \in [0, U]$ at fee $2nf$. Any larger fleet has the same upper endpoint U and a higher fee when $f > 0$. It is therefore dominated at each feasible load. The literal projection acquires higher-cost layers, but its lower frontier and support conclusions are unchanged. This argument does not apply when $H > S$: additional idle inventory can then enable negative net early loads and may be useful.

B.2 Persistence and removal of an intermediate charge requirement

For the earlier anchor with A energies $(5, 15)$, B energies $(14, 15)$, battery capacity 20, $K = 15$ and $H = 49$, the matching values are $(r_0, r_1, r_2) = (0, 0, 9)$ and the new thresholds remain

$$(t_0, t_1, t_2) = (0, 0, 9).$$

Although individual buses gain discharging options, the entire cost-labelled projection is unchanged. At the earlier fee and supply objective, its previous positive gap therefore survives analytically. The terminal bound here enforces $E \ge 0$, so this particular conclusion does not require evaluating the old

supply function at negative fleet loads. This is a proof consequence, not a new V2G implementation qualification or experiment.

Bidirectional charging can also remove a positive gap. Take complete crossmatching, A energies $(1, 1, 19)$, B energies $(2, 10, 19)$, capacity 20, $K = 2$, $H = 52$ and fee $f = 1$. Here $S = 52$ and $(r_0, r_1, r_2, r_3) = (0, 0, 0, 1)$. In the charge-only model, the cheapest physical operating cost is 4 on $0 \le E < 1$ and 3 on $1 \le E \le 2$. Its hull boundary is $4 - E$ on $[0, 1]$, followed by the constant 3. For the convex supply cost

$$F(E, L) = E^2 + \frac{E + L}{5}, \qquad G(E) = E^2 + \frac{52}{5},$$

the physical minima at $E = 0$ and $E = 1$ tie, whereas the hull minimum is at $E = 1/2$. Consequently,

$$D = \frac{72}{5}, \qquad \mathrm{CH} = \frac{283}{20}, \qquad D - \mathrm{CH} = \frac{1}{4}.$$

With bidirectional charging, all four thresholds t_m are zero, so the three-bus operating cost 3 is physical throughout $[0, 2]$. In particular, pair the services as $(1, 19)$, $(1, 10)$ and $(19, 2)$ and assign signed early charges $0, -1, 1$, respectively. Their net load is zero and gross throughput is 2. Their terminal refills are $20, 12, 20$, totaling 52; all intermediate inventories respect capacity. Thus

$$D = \mathrm{CH} = 3 + \frac{52}{5} = \frac{67}{5}, \qquad E^* = 0.$$

At the own-load price $\nabla F(0, 52) = (1/5, 1/5)$, every feasible plan pays $52/5$ for net energy, and the three-bus plan minimizes operating cost. Hence it is supported under the stated settlement rule. Discharge from another assigned bus offsets a mandatory positive early purchase, using the gross connector budget in both directions. These exact values follow analytically from the projection theorem; they are not a numerical implementation qualification.

Losses cannot be incorporated by rescaling the early-energy threshold alone. Let η_c, η_d denote charge and discharge efficiencies, X early grid exports and E early net grid purchase. Full replenishment requires terminal grid purchase

$$L = \frac{S}{\eta_c} - E + \left(\frac{1}{\eta_c \eta_d} - 1\right) X.$$

For strict round-trip loss, the same net early load can therefore have different terminal demand. The lossless one-dimensional projection does not transfer directly. Under the common-window assumptions and nonnegative prices or marginal supply cost, within-bus cycles can be removed without worsening the objective, but cross-bus redistribution cannot generally be removed in that way. Section C derives a joint efficient-frontier reduction with common losses; the lossless scalar support classification does not follow from that reduction.

The result concerns lossless common-window physics and the specified net settlement. It neither establishes novelty of V2G or matching theory nor covers degradation, asymmetric prices, realistic travel, or the public 37-service case. That case remains charge-only and unchanged. No numerical qualification of this bidirectional extension is claimed.

C Common losses and the joint load–cost set

With losses, early net grid energy no longer determines terminal purchases. The same cardinality-matching values nevertheless suffice under the common-window assumptions. Retaining finite cycling actions gives a complete polygonal projection. Discarding dominated actions gives a smaller

efficient frontier for monotone economic objectives. The two descriptions have different economic and physical qualifications.

Retain the services, allowed graph, battery capacity B, positive fee f and service-assigned buses of Section B. Every bus starts and finishes full, has zero reserve, and is available throughout the early window. There is no travel, auxiliary consumption, taper or switching time. Let $\alpha \in (0,1]$ be the common charging efficiency in both windows and $\delta \in (0,1]$ the common discharging efficiency. Powers and energies are measured on the *grid side*: early purchases C and exports X use the shared connector's gross budget $C + X \le K$, and terminal grid purchases satisfy $L \le H$. Each bus accepts the corresponding connector power. Signed exports receive the same posted rate as purchases. Write

$$S = \sum_i a_i + \sum_j b_j, \qquad P = \sum_i a_i, \qquad \kappa = \frac{1}{\alpha\delta} - 1 \ge 0.$$

Battery exchange is $\alpha C_v - X_v/\delta$, whereas net early grid energy is $E = C - X$. Full replenishment gives

$$L = \frac{S}{\alpha} - E + \kappa X. \tag{11}$$

Thus two allocations with the same E may have different L. The supply objective must be defined on the resulting signed load domain, or on an explicitly declared background-demand translation.

C.1 The complete projection when finite cycling is allowed

An exact projection can retain the otherwise dominated cycling actions. We now explicitly permit any finite number of early charge/discharge actions on a bus. There is no switching time, minimum session, separate throughput fee, degradation cost or cycle limit. Common connector powers are admissible in both directions for every bus, and all buses remain available throughout the early window. Assume $B > 0$ and a nonempty used fleet. Economic objectives in this subsection depend on period-aggregate loads and vehicle cost, rather than the timing or number of switches. These assumptions are essential to the full-set claim.

Theorem 12 (Complete common-loss load–cost projection)**.** *For matching cardinality $m = 0, \dots, \nu$, let $N_m = 2n - m$ and let r_m be the least total early battery requirement among allowed matchings of size m. Define the compact polygon, possibly empty,*

$$\begin{aligned} W_m = \{(C,X) : C \ge r_m/\alpha, \quad X \ge 0, \quad C + X \le K, \\ \max\{S - N_m B, S - \alpha H\} \le Q = \alpha C - X/\delta \le P\}. \end{aligned} \tag{12}$$

The complete physical load–cost projection is exactly

$$\bigcup_{m=0}^{\nu} \left\{ \left(C - X, \frac{S - Q}{\alpha}, f N_m \right) : (C,X) \in W_m \right\}.$$

Every point is attained by a finite-action schedule. Thus arbitrary economic objective values on these projected quantities, including negative posted-price value queries, are preserved without a monotonicity assumption. Each nonempty polygon has at most five vertices; their affine images generate the complete-plan convex hull with at most $5(\nu + 1)$ records, including lossless or lower-dimensional degeneracies.

Proof. Fix one matching, with fleet N, summed requirement R and $T = S - NB$. Let x_v be a bus's net early battery increment. Its paired, A-only and B-only endpoint boxes are respectively

$$[a_i + b_j - B, a_i], \qquad [a_i - B, a_i], \qquad [b_j - B, 0].$$

Their lower endpoints sum to T, their positive lower endpoints to R, and their upper endpoints to P. Every upper endpoint is nonnegative. Regardless of action order, $x_v = \alpha C_v - X_v/\delta$, so total $Q \in [T, P]$. A positive lower endpoint requires at least its value divided by α in purchases. Hence $C \geq R/\alpha$ is necessary. The other inequalities are nonnegative exports, shared gross use and full-terminal accounting $L = (S - Q)/\alpha \leq H$. Terminal purchases are nonnegative because $Q \leq P < S$; individually each box upper endpoint also leaves a nonnegative refill requirement.

Conversely fix aggregate totals satisfying these inequalities with R in place of r_m. For any $Q \in [T, P]$, the boxes admit an exchange vector whose positive parts sum to $A = \max\{R, Q\}$. If $Q \geq R$, start from positive lower endpoints and zero elsewhere, distributing $Q - R$ through the remaining headrooms. If $Q < R$, keep mandatory positive increments and distribute negative increments of total $Q - R$ through the negative lower intervals, whose combined lower endpoint is $T - R$. One-direction exchanges then realize

$$C_0 = A/\alpha, \qquad X_0 = \delta(A - Q).$$

Since $X \geq 0$ implies $\alpha C \geq Q$, and $\alpha C \geq R$, we have $z = C - C_0 \geq 0$ and $X - X_0 = \alpha\delta z$.

Realize these residuals by battery-neutral cycles on one existing bus before its one-direction exchange. At early SOC $s \in [0, B]$, use charge-then-discharge cycles of depth at most $B - s$ if $s \leq B/2$, and discharge-then-charge cycles of depth at most s otherwise. The permitted depth is at least $B/2 > 0$. Split the residual battery throughput αz into finitely many such chunks, allowing a smaller last chunk. Each returns to s, stays in $[0, B]$, and uses grid purchases depth$/\alpha$ and exports δ times depth. Their total residual is exactly $(z, \alpha\delta z)$, with no limiting or Zeno argument.

Execute the cycles and then all one-direction exchanges serially. Their gross grid throughput is exactly $C + X \leq K$, so they fit the common early window at the admitted power. Endpoint boxes ensure subsequent service feasibility; nonnegative service consumption preserves intermediate SOC. Refill serially at the terminal, using total $L \leq H$. If $K = 0$, all early actions are zero and no positive-power division is required. If $H = 0$, only zero terminal amounts could be admitted; under the present positive B-service energies, $P < S$ makes every polygon empty, consistently with physical infeasibility.

At fixed m, only R depends on the matching: S, P, N, T and the fee are constant. Its entire polygon is contained in the polygon for a matching attaining r_m, and every point of that larger polygon has just been physically constructed. This proves exact equality of the union, not merely economic dominance. Equation (12) has five halfspaces in the plane. Its bounded polygon has at most five vertices; affine images and then finite convex combinations prove the generator claim. When $\alpha\delta = 1$, the map collapses to $E + L = S/\alpha$ rather than being invertible; the same argument still applies. □

Corollary 13 (Capped identical idle buses in the complete projection)**.** *Under the same zero-transition model, permitting identical paid idle buses with a total fleet cap* $2n$ *leaves this complete projection unchanged.*

An idle bus has exchange box $[-B, 0]$. A fleet with matching size m and q idle buses has $q \leq m$ by the cap. Delete q matching edges and serve their services separately. This preserves total N, S, P, T and fee, while lowering R. The resulting all-assigned polygon reproduces the same C, X, Q, E, L, c, including cycles. Reverse inclusion is immediate. Identical access, efficiency, capacity and fees

are required; nonzero transition consumption would change S under deletion and needs a separate argument.

Let V be the vertex-image records in the theorem. An exact continuous formulation for complete-plan convexification is the familiar relation

$$\operatorname{conv} \mathcal{P} = \left\{ \sum_{v \in V} \lambda_v v : \lambda_v \geq 0,\ \sum_{v \in V} \lambda_v = 1 \right\},$$

where $\mathcal{P}$ denotes the complete projected set above. A convex F gives ordinary convex planning on each polygon and on this hull; physical planning takes the minimum over the cardinalities. These are standard convex-optimization tools. Merely interpolating r_m at a fractional cardinality and substituting into physical inequalities is not this disjunctive hull construction and remains unjustified.

Removing a neutral cycle keeps Q, L fixed and lowers E by $(1 - \alpha\delta)z$. This explains why monotone objectives can discard it in the efficient-frontier result below, while arbitrary objectives must retain the full polygons. Its smaller set Z still requires the stated monotonicity assumptions. The finite cycle count can be large; no runtime or generic hull-algorithm claim is made. A separate eight-case exact qualification of these full polygons, described in Appendix Y, includes finite-action physical replay. It supplies no public-timetable V2G, empirical or learning evidence. The earlier one-direction qualification has a different scope.

Why a negative-price query needs the full set. Take $n = 1$, $B = 1$, $a = b = 3/4$, $\alpha = 1/2$, $\delta = 1$, $K = 2$, $H = 3$ and $f = 2$. At the fixed posted prices $p = (-3, 1)$, the private score is $c - 3E + L$. For the paired one-bus assignment,

$$C = 11/6, \quad X = 1/6, \quad Q = 3/4, \qquad (E, L, c) = (5/3, 3/2, 2)$$

gives score $-3/2$. This point has a finite physical realization. After A the battery holds $1/4$: purchase $1/3$ grid units and export $1/6$, returning to $1/4$, then purchase $3/2$ to reach full. These three early actions have gross energy 2 and serialize at power 2 within the early window. B leaves $1/4$; terminal purchases $3/2$ restore full inventory within the power-3 terminal window.

This is the exact response value, not just a feasible score. For $m = 1$, the polygon has $C \geq 1$ and $1/2 \leq C/2 - X \leq 3/4$; for $m = 0$, it has $C \geq 0$ and $0 \leq C/2 - X \leq 3/4$. Both impose $X \geq 0$ and $C + X \leq 2$. Their scores are respectively $5 - 4C + 5X$ and $7 - 4C + 5X$. For any feasible C, use the smallest export $X = \max\{0, C/2 - 3/4\}$. The resulting score decreases on both pieces, with final endpoint $C = 11/6$, giving minima $-3/2$ and $1/2$, respectively. Thus the complete-fleet response value is $-3/2$.

The efficient curves instead have $L = 3 - E$: the two-bus curve has $0 \leq E \leq 3/2$, $c = 4$, and the one-bus curve has $1 \leq E \leq 3/2$, $c = 2$. Their best scores are 1 and -1. Discarding the neutral cycle therefore changes this query's value from $-3/2$ to -1. It raises early purchases without changing terminal refill, so the negative early rate rewards a point that monotone economics discards. Because the queried objective is linear, its minimum over the physical set and its complete-plan hull is the same: $D = \mathrm{CH} = -3/2$ for this objective. This is not a positive planning gap or an equilibrium claim. This analytical anchor also passed the separate full-polygon implementation control; it gives no public-timetable V2G conclusion.

Corollary 14 (Directional powers, affine wear and a finite action cap)**.** *Retain the full-action assumptions of Theorem 12. In the unit early window let common purchase power be $K_c > 0$ and*

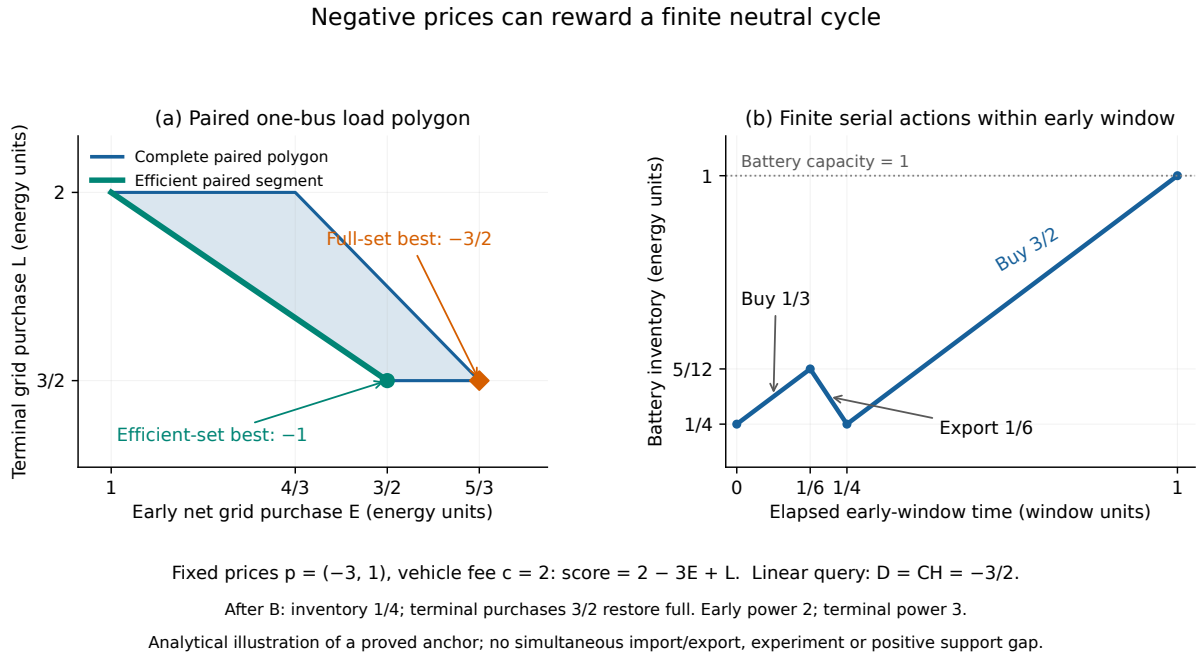


Figure 1: The full paired polygon retains a finite neutral cycle that the efficient frontier discards. At the negative posted price, that cycle improves the fleet's linear response value. This illustration has $D = \mathrm{CH}$; it is not a positive planning-gap example. The right panel shows the three serial early actions of the same physical witness.

export power $K_d > 0$, accepted by every bus in the corresponding direction. Replace the gross-budget halfspace by

$$C/K_c + X/K_d \le 1.$$

With the other four halfspaces unchanged, these are still the exact cardinality polygons. For common affine grid-throughput wear coefficients, their complete joint load–cost images are

$$E = C - X, \qquad L = (S - Q)/\alpha, \qquad c = fN_m + w_c C + w_d X + w_t L.$$

Each polygon still has at most five vertices, whose affine images generate the complete-plan hull. Fixed nonnegative wear coefficients have their usual cost interpretation; exact projection equality itself needs no coefficient-sign or economic-monotonicity restriction.

For a fixed used fleet N, every attainable aggregate point has a realization with at most

$$N + 2\left\lceil \frac{2\alpha K_c}{B} \right\rceil \quad \textit{early charge/export actions}, \qquad N \quad \textit{terminal charge actions}.$$

An action cap at least these bounds preserves the aggregate projection, not every literal schedule. Replacing N by $2n$ gives one bound for all cardinalities under the standing fleet cap.

Proof. The endpoint boxes, mandatory purchases and terminal accounting are unchanged. Necessity of the new halfspace follows because the two directions occupy at least C/K_c and X/K_d connector time and cannot overlap. For sufficiency use the same baseline exchanges and neutral cycles as in Theorem 12. Execute each purchase at power K_c and each export at K_d; their total serial duration is exactly the displayed occupancy. Directional asymmetry adds no SOC condition.

At fixed cardinality a minimum-R matching reproduces the same C, X, Q, L, hence the same affine wear cost, for any other matching. This proves complete-set equality; affine imaging and convex combination give the vertex and hull claims. The common-task version uses S_m in place of S, as in Corollary 19.

The residual $z = C - C_0$ has battery throughput $\alpha z \le \alpha K_c$. Choosing chunks of depth $B/2$, with a possibly smaller last chunk, needs at most $J = \lceil 2\alpha K_c/B \rceil$ cycles, each with two actions. The chosen orientation fits the bus's initial SOC, and each cycle returns to it. Each bus then needs at most one baseline early action and one terminal charge; zero actions are omitted. Restricting to this action count adds no aggregate point and removes none, since every point has the explicit bounded-action realization. The count is conservative, not minimal, and does not bound runtime or the number of actions in arbitrary schedules. □

Wear must remain in the cost image. In the lossless case a neutral cycle adds equal amounts z to C, X, leaving E, L unchanged but adding $(w_c + w_d)z$ to intrinsic cost. Thus net loads and a constant vehicle fee alone cannot represent this complete joint projection. For the same reason, convexification retains each component's actual wear cost. Affine wear commutes with averaging when C, X, L remain separate; mean net loads alone need not determine it.

This corollary does not transfer the smaller efficient-Z formulas to unequal powers or wear, nor cover SOC/depth/rate-dependent degradation, per-session cost, positive switching time, minimum durations or heterogeneous acceptance. Its positive-power assumptions are separate from the theorem's zero-budget cases. It is an analytical extension with no numerical implementation qualification: the eight full-polygon cases use symmetric early powers and no wear. No public-timetable, empirical wear-calibration or computational-advantage conclusion follows.

C.2 The smaller efficient frontier for monotone costs

Lemma 15 (Least terminal purchase at a fixed early net load)**.** *Fix an allowed matching M of size m, using $N = 2n - m$ buses. Put*

$$R = \sum_{(i,j)\in M} [a_i + b_j - B]_+, \quad T = S - NB, \quad d_R(E) = \left[\frac{R}{\alpha} - E\right]_+, \quad L_R(E) = \frac{S}{\alpha} - E + \kappa d_R(E).$$

Among early allocations with at most one direction of action per bus, the feasible early-net interval is exactly $[t(R,N), U]$, when nonempty, where

$$U = \min\{P/\alpha, K\}, \tag{13}$$

$$t(R,N) = \max\left\{(1/\alpha - \delta)R + \delta(S - NB), \quad 2R/\alpha - K, \quad S/\alpha - H, \quad \frac{S/\alpha + \kappa R/\alpha - H}{1+\kappa}\right\}. \tag{14}$$

For each such E, the least attainable terminal purchase is $L_R(E)$. It is achieved with $C = E + d_R(E)$ and $X = d_R(E)$.

The first threshold in (14) accounts for available battery inventory; the second prevents offsetting purchases and exports from evading connector time. The last two enforce terminal refill on both pieces of L_R. In particular, replacing service energy by S/α in the lossless formula would omit both the export loss and its connector consequences.

Retain the allowed-graph matching values r_m from Section A, for $m = 0, \ldots, \nu$. Define $N_m = 2n - m$, $t_m = t(r_m, N_m)$, $L_m = L_{r_m}$ and $c_m = fN_m$.

Theorem 16 (Finite efficient-frontier reduction with common losses)**.** *The set*

$$Z = \bigcup_{\substack{0\le m\le \nu \\ t_m \le U}} \{(E, L_m(E), c_m) : t_m \le E \le U\} \tag{15}$$

is physical. Every physical point is weakly dominated coordinatewise in (E, L, c) *by a point of* Z *with the same operating cost. Hence the complete physical set's coordinatewise efficient frontier is obtained by removing dominated points from* Z*. The thresholds* t_m *are nondecreasing and discretely convex.*

Each cardinality contributes at most two line segments, with a possible kink at $E = r_m/\alpha$*. Thus* $\operatorname{conv} Z$ *is the convex hull of finitely many segment endpoints and kinks. If the convex supply objective is coordinatewise nondecreasing on the admitted signed domain, physical and hull planning values can be computed over* Z *and* $\operatorname{conv} Z$*, respectively, without changing either value.*

Dominance here allows a replacement with smaller early load as well as smaller terminal load; it does not identify the full physical set. Under nonnegative early and terminal posted prices, the same reduction preserves the minimum bill plus operating cost. At zero prices dominated allocations can tie, so it does not enumerate every price response. Negative prices or a nonmonotone supply objective may require the alternatives retained by Theorem 12; they cannot use the smaller set Z by assertion. Asymmetric export settlement must keep its distinct purchase/export coordinates. The lossless scalar criterion for the location of a supported hull optimum is not asserted here: different cardinalities can have different terminal loads at the same E. Convex network dispatch with increasing generation costs need not give a load-monotone value function: congestion can make a nodal marginal price negative. Appendix U gives an exact illustration and the established pricing precedent. The smaller efficient set therefore requires a load-domain monotonicity check. The complete polygons instead retain all period-aggregate load/cost points, subject to their explicit cycling assumptions and the network objective's declared domain.

Identical paid idle buses remain redundant for these efficient values under the total fleet cap $2n$. Deleting as many matching edges as there are idle buses preserves the total fleet and fee and lowers R, so an all-assigned fleet has no larger L at the same E. This efficient-set argument gives dominance. If finite cycling is retained, the complete projection result above gives the stronger equality of the whole projected set, as its capped-idle corollary explains. There is no general uncapped-idle conclusion when negative net early loads are valuable. The theorem remains conditional on common efficiencies, common windows and ideal serial use; it is not a public-timetable V2G result. A separate ten-case exact qualification matched these efficient-frontier formulas to independent one-direction SOC/sign-polytope enumeration, checking all vertices of those sign polytopes and attainment along complete frontier segments. It did not qualify the full cycling polygons. These finite implementation checks supplement the proof; they do not establish empirical validity or prevalence.

C.3 Planning and support from the finite frontier

The reduction gives a direct way to compute the two planning values under the monotonicity assumptions of Theorem 16. There are at most $2(\nu + 1)$ affine segments and $M \le 3(\nu + 1)$ generating records $v_j = (u_j, c_j)$, where $u_j = (E_j, L_j)$; empty intervals are omitted and singletons require only one record. Physical planning minimizes $c_m + F(E, L_m(E))$ on each segment, a convex one-dimensional problem, and takes the smallest value. Hull planning solves

$$\min_{\lambda\ge 0,\ \sum_j \lambda_j = 1} \left\{ \sum_j \lambda_j c_j + F\left(\sum_j \lambda_j u_j \right) \right\}.$$

These use standard matching and convex-optimization tools; the physical result is that the matching values suffice to construct their input.

Remark 17 (Interpolate complete plans, not fractional fleet physics). Interpolating r_m and substituting a continuous cardinality into (14) need not give this hull. Take one service per batch, $B = 1$, $a = b = 3/4$, $\alpha = 1/2$, $\delta = 1$, $K = 2$, $H = 3$ and $f = 2$. Then $r_0 = 0$, $r_1 = 1/2$; the two complete efficient branches are

$$(E, 3 - E, 4), \quad 0 \le E \le 3/2; \qquad (E, 3 - E, 2), \quad 1 \le E \le 3/2.$$

Both physical and hull minima of $c + 3E + L$ equal 7. Instead put $r(m) = m/2$, $N = 2 - m$ and $c = 2N$ for fractional m. The same formulas admit $m = 1/2$, $E = 1/4$, $L = 3$, $c = 3$, of cost $27/4 < 7$: the four lower bounds in (14) are $1/4, -1, 0, 1/4$, and the terminal formula gives 3. Every actual mixture with mean cardinality m requires $E \ge m$, so this point is fictitious. Taking maxima of resource bounds after averaging can relax the averages of complete feasible branches. Convexity of the matching sequence proves neither equality nor physical attainability of that shortcut. The finite generator formulation above retains the required coupling exactly. This example has no true physical–hull gap: the artificial reduction creates one.

For differentiable F, a proposed hull point $(\bar{u}, \bar{c}) = \sum_j \lambda_j (u_j, c_j)$ has a short sufficient certificate: feasible weights, its exact objective and gradient $p = \nabla F(\bar{u})$, and the inequalities

$$c_j + p \cdot u_j \ge \bar{c} + p \cdot \bar{u} \qquad \text{for every generating record } j.$$

Convexity proves optimality on the hull; nonnegative p and physical dominance extend the private-score lower bound to all original plans. For any nonnegative posted price, the minimum private score is likewise the minimum over these records. At zero prices this computes the value, not every response: discarded dominated plans or entire segments can tie.

If the generators and convex quadratic coefficients are rational, there exists a hull optimum with rational loads, cost and weights, permitting an exact rational certificate. Rational physical data, including the efficiencies, suffice for rational generators; a rational objective alone does not. At any optimum the supporting functional has normal $(p, 1)$, so its face has dimension at most two, or the whole hull already does. Thus at most three generators suffice. Enumerating their simplex faces and solving rational linear stationarity systems also handles singular positive-semidefinite quadratics; no inverse of a singular Hessian is required. Flat optimum sets can contain irrational points despite the existence of a rational representative. These finite deductions make no new runtime or general scheduling-complexity claim.

D Which transition energies preserve the reduction?

Zero transition energy ensures that total service consumption is independent of the matching. A controlled extension permits a common energy cost per matched pair. This is a compulsory service-transition task, performed over a common positive-duration, noncharging interval after the early window and before the B batch. It is an abstract extension of the common-window model, not a claim about arbitrary road movements or charging access.

Corollary 18 (Common transition energy)**.** *Retain the assumptions of Theorem 16, except that each matched A–B pair consumes the same additional energy $d \ge 0$ during that interval, with nonnegative consumption throughout. Insert this interval after early exchange and translate the B batch and terminal window later without changing their durations or budgets: for example, A*

on $[0,1]$*, early exchange on* $[1,2]$*, transition on* $[2,3]$*,* B *on* $[3,4]$ *and terminal exchange on* $[4,5]$*. Singletons perform no transition task. Restrict the allowed graph to edges with* $b_j + d \le B$*. Write* $S = \sum_i a_i + \sum_j b_j$*,* $P = \sum_i a_i$*,* $S_m = S + md$*,* $N_m = 2n - m$*, and*

$$r_m = \min_{|M|=m} \sum_{(i,j)\in M} (a_i + b_j + d - B)_+, \quad U = \min\{P/\alpha, K\}, \quad \kappa = \frac{1}{\alpha\delta} - 1.$$

For $m = 0, \dots, \nu$ *in this restricted graph, the efficient curve is*

$$L_m(E) = \frac{S_m}{\alpha} - E + \kappa \left(\frac{r_m}{\alpha} - E \right)_+, \qquad t_m \le E \le U,$$

where

$$t_m = \max\Bigg\{ (\alpha^{-1} - \delta) r_m + \delta(S_m - N_m B), \quad 2r_m/\alpha - K, \\ S_m/\alpha - H, \quad \frac{S_m/\alpha + \kappa r_m/\alpha - H}{1+\kappa} \Bigg\}.$$

Intervals with $t_m > U$ *are infeasible. There are at most two efficient segments per cardinality. Under coordinatewise nondecreasing supply costs, these curves suffice for physical and convexified planning values and for nonnegative posted-price value queries, as in Section C.3. They do not enumerate every response tie or the full physical projection. The sequence* t_m *is discrete convex.*

The proof uses the same interval allocation as Lemma 15: a paired bus now has early battery-exchange box $[a_i + b_j + d - B, a_i]$. Its lower endpoints sum to $S_m - N_m B$ and its positive lower endpoints to the matching requirement. Thus both the requirement and the terminal energy account include the task. Appendix V gives the details, including capped paid-idle dominance. No lossless scalar support criterion is transferred by this corollary. The separate full-polygon qualification includes one common-task literal and checks the efficient lower threshold as well; the continuous-domain claim rests on the proof.

Corollary 19 (Complete polygons with a common transition task)**.** *Retain the timing, restricted graph* $b_j + d \le B$ *and definitions* S_m, N_m, r_m, P *of Corollary 18, and permit the finite early actions of Theorem 12. For service-assigned fleets, the complete physical projection is the union over* $m = 0, \dots, \nu$ *of*

$$\left\{ \left(C - X, \frac{S_m - Q}{\alpha}, f N_m \right) : (C, X) \in W_m^d \right\},$$

where

$$W_m^d = \{(C,X) : C \ge r_m/\alpha, \quad X \ge 0, \quad C + X \le K, \\ \max\{S_m - N_m B, S_m - \alpha H\} \le Q = \alpha C - X/\delta \le P\}.$$

Each nonempty polygon has at most five vertices. Their affine images generate the complete-plan hull. Keeping these full polygons preserves arbitrary projected-objective values, including negative-price queries; no monotonicity assumption is required.

Proof. The paired exchange box is $[a_i + b_j + d - B, a_i]$; singleton boxes are unchanged. Their lower endpoints sum to $S_m - N_m B$, upper endpoints to P, and positive lower endpoints to the matching requirement. The finite-cycle construction of Theorem 12 occurs before the transition task and returns to the starting early SOC. Nonnegative task and B-service consumption then preserves intermediate SOC feasibility. At fixed m, S_m is independent of the matching, so a matching attaining r_m contains every other matching's polygon. The full-set and hull conclusions follow. □

For $d > 0$ this is not a literal paid-idle elimination result. Deleting an edge while replacing its paired bus and an idle bus by two singletons reduces total consumption by d. It changes terminal load even at the same C, X, so only the earlier monotone-value dominance claim follows. The statement also excludes edge-dependent task energies, charging during the task and altered access policies. One common-task literal passed the separate full-polygon qualification; it does not qualify arbitrary task models.

Varying edge energies need more information. The failure already occurs with charging only; disable early exports in this example. Consider $n = 2$, $a = b = (13, 1)$, $B = 20$, unit efficiency, early budget $K = 14$ and terminal budget $H = 40$. Every edge is allowed, with transition energies $\left(\begin{smallmatrix} 0 & 6 \\ 6 & 0 \end{smallmatrix}\right)$. Use A services on $[0, 1]$, early charging on $[1, 2]$, positive-duration noncharging transitions on $[2, 3]$, B services on $[3, 4]$ and terminal charging on $[4, 5]$. One connector of power 14 early and 40 terminal, with adequate acceptance, serially realizes the stated budgets. Access and other nonservice energy are zero. Transition tasks are compulsory; no metric-road realization or substitute route is assumed.

Full matching	Minimum early requirement	Total energy	Load segment
Identity	6	28	$(E, 28 - E),\ 6 \le E \le 14$
Cross	0	40	$(E, 40 - E),\ 0 \le E \le 14$

At $E = 6$ the identity's terminal load 22 is smaller than every cross load with early purchase at most 6. Selecting only the minimum-requirement matching therefore loses an efficient branch. For the fixed monotone convex supply cost $F(E, L) = E + L$, its two-bus value is $2f + 28$ instead of the cross value $2f + 40$. Allowing singleton duties does not repair the error when $f > 0$: the true value is $2f + 28$, whereas even the best minimum-requirement choice at every cardinality gives

$$\min\{2f + 40, 3f + 28, 4f + 28\} = 2f + 28 + \min\{12, f\} > 2f + 28.$$

This elementary counterexample isolates the lost assignment-independence of total energy. It is neither a general hardness result nor a public-case or empirical finding. Appendix W extends the construction to 2^k efficient physical pieces at a single fleet size, each necessary for some monotone convex planning objective. Its numerical magnitudes grow with k, and the hull remains simple; it proves neither hardness nor an exponential requirement for every symbolic representation. For one smooth monotone convex objective in that family, the exact gap is $9/(2K) > 0$ even with all these alternatives available. The gap decreases with the family's charging budget K; there is no uniform-size claim. Appendix X then adds every interblock edge and admits every perfect matching, while preserving that gap. Thus graph disconnection is unnecessary for this example. The compulsory-task model and fleet cap remain essential stated restrictions.

E An exact positive gap on one public timetable

For the stated charge-only movement-graph model of one public 37-service timetable, an independently checked physical lower bound strictly exceeds a feasible complete-plan mixture's hull upper. The outward bounds are

$$\mathrm{CH}_{15} \le 441.142238 < 441.532912 \le D_{15} \le 441.532913, \qquad D_{15} - \mathrm{CH}_{15} \ge 0.390674. \tag{16}$$

The gap lower is the conservative difference of the down-rounded physical lower and up-rounded hull upper shown here; it is not an exact gap value. The upper on D_{15} rounds upward the qualified

exact witness value $441.532912\ldots$. Neither D_{15} nor CH_{15} is known exactly. By Proposition 2, no physical fleet is supported at its own marginal prices under this declared model and tariff convention. This is an exact sign certificate with a conservative separation; it is not evidence of a practically large loss or operational prevalence.

E.1 The declared case and objective

The case uses the Hildenbrand timetable from the Sistig data [19, 20]: 37 mandatory services and the complete directed deadhead matrix. Depot 15 and depot 16 are separate one-depot modifications of the same base timetable, not independent operational samples. The nonlinear sign certificate concerns depot 15. Source timetable offsets, including next-day services, are retained; the source data do not supply the energy or economic calibration below.

The opportunity set is the explicitly declared finite movement graph and its descriptor's fleet cap. Its pullout, direct service-to-service, via-depot and pullin modes have fixed leg times and specified depot charging gaps. Services may be assigned to any compatible complete path in this graph; charging allocations within the declared gaps remain continuous and share the connector. The complete directed deadhead matrix supplies movement data, but does not by itself authorize arbitrary retiming, extra depot visits or unlisted movement modes. The certificate quantifies over all fleets admitted by this graph. Extending that claim to a larger continuous-routing opportunity set requires an additional coverage or dominance argument, not merely the same timetable.

Two separate exact metadata checks narrow this limitation. The source's five-location directed time and distance tables satisfy every triangle inequality. Independent comparisons also verified all 37 service mappings, all 1370 emitted modes and their 2648 legs against the certified descriptor, including every required or time-infeasible direct and via-depot incidence. Thus the declared graph is complete for the stated canonical constructor. These checks establish correspondence and stored-energy provenance, not energy dominance over arbitrary additional walks or continuously retimed waiting. In particular, small coefficient differences can cross battery or charging-feasibility thresholds; a positive objective gap alone does not justify changing the energy law. The declared-graph scope is retained. Section F defines a separate exact-consumption model and obtains fresh certificates before extending its scope.

Service consumption is 1.20 kWh/km plus 16 kWh per service hour; directed deadheads use 0.96 kWh/km plus 16 kWh per travel hour, and explicit off-depot waiting uses 16 kWh/h. Depot-parked auxiliary draw is zero. Each used bus has 400 kWh usable inventory, zero reserve, unit efficiency and 360 kW acceptance. One 360 kW connector is shared. Every used bus starts full and returns full by a synthetic 30:00 service-block boundary. There is no taper or switching time. Monetary movement cost is zero, but movement time and energy remain physical constraints. The used-bus fee is 100. This finite block is not a certificate of repeating daily operation. The public model remains charge-only; the preceding V2G classification is a separate analytical model and was not applied to this timetable. Service, movement and waiting energies use the original stored binary64 values produced in the source's operation order; the formulas above describe their provenance, not a replacement by unrounded rational coefficients.

For its 30 hourly market intervals, the convex supply objective is

$$F(e) = \sum_{t=1}^{30} \left(0.2 e_t + \frac{e_t^2}{1800} \right). \tag{17}$$

These familiar decimal coefficients denote the original stored binary values used in every exact

certificate.[2] The coefficients are synthetic. The result therefore concerns a specified ideal scheduling and supply-cost instance, not calibrated market behavior, a strategic pricing equilibrium or an ML experiment.

E.2 What establishes the strict separation

The physical upper is a rational charging allocation on selected routes, independently replayed against every service, movement, battery bound, shared connector and full terminal inventory. A separate rational dual certificate establishes its charging optimum conditional on those routes, not global fleet optimality.

A different strictly admitted physical response has a higher nonlinear planning cost but a lower bill at this incumbent's fixed marginal prices. It therefore supplies a genuine deviation, rather than a raw floating candidate. An exact two-plan mixture of these admitted plans, checked by direct quadratic evaluation and one-dimensional optimality conditions, supplies the hull upper in (16). A mean complete-plan load is not a physical daily dispatch; the mixture upper is not asserted to be the global hull optimum.

The missing global step is supplied by a new physical branch proof. An independently reconstructed minimum-fleet obstruction rules out one bus within the declared movement graph. For every fleet with at least three buses, the vehicle fee and only the linear mandatory-service-energy term already exceed 477.82; omitted movement consumption and quadratic supply cost are nonnegative. These facts permit searching the two-bus physical domain while covering every other fleet count separately. They do not exclude larger-fleet components from arbitrary hull mixtures or from response problems at other tariffs.

Binary branching retains both values of each selected movement variable. For every fringe region, exact tangent planes, signed row multipliers and finite-box residual compensation give a valid lower; an infeasible region requires an exact positive contradiction. Independent semantic reconstruction and complete branch coverage make these bounds exhaustive. A bounded refinement retains the original partition and strengthens its leaves under their inherited decisions, yielding (16). The outward physical interval has width at most 10^{-6}, but equality of the exact bounds is not claimed. Appendix R gives the checkable algebra and Appendix R.2 records the search and qualification stages. This is a global physical lower, not a hull lower; a native OPTIMAL status, finite plan bank or small solver gap would not establish it.

E.3 Stability to the vehicle fee

Holding physics and the supply objective fixed, every used-bus fee $f \geq 100$ retains the positive gap guarantee in (16). Both admitted mixture components use two buses, so their fee shift cancels that of the two-bus lower; a separate nonnegative-fee floor covers larger fleets. An independently checked exact interval calculation also gives an open sufficient range below 100. Appendix R.1 gives the derivation and endpoint qualifications. This varies one economic parameter, not losses, battery capacity or station resources.

E.4 Interpretation and limitations

Classical native planning already found an objective near 441.532912; the contribution of the certificate chain is exact admission and exhaustive coverage, not a better heuristic objective or an ML

[2]The linear coefficient is 3602879701896397/18014398509481984 and the quadratic Hessian coefficient is 5124095576030431/4611686018427387904; the quadratic term is half the latter coefficient times e_t^2.

advantage. Earlier failed physical screens and multiplier recovery remain failures in Table 1; later certificates do not relabel them. The earlier continuous relaxation's valid hull lower 434.17 remains distinct from the branched physical lower. Appendix R.2 also separates flat-tariff comparisons, internal qualifications and the new portable retained-evidence check. The physical value is tightly enclosed, but the hull optimum and exact gap remain unknown. One timetable with synthetic energy and economic assumptions cannot estimate prevalence, operating savings or welfare effects in bus systems. The result extends the analytical possibility to this fully specified varied-time charge-only instance; it does not validate the simplified common-window model empirically or establish a learning contribution.

F A conditional extension to exact additive consumption

The public result in Section E uses the original stored binary consumption coefficients. It does not automatically cover a change to unrounded energy formulas. Here we state a separate, conditional model extension with a separately checked positive-gap certificate. No physical bound or feasible point from (16) is assumed to survive the coefficient change. The argument is a classical triangle-inequality and charging-curtailment argument, not a new generic routing result.

The new law and opportunity set. Each ordered pair of a fixed finite location set is an available primitive arc with jointly specified time τ_{ij} and distance d_{ij}. These are abstract modeled pairs; their simultaneous realization by an empirical road path is not inferred. Assume both directed tables are nonnegative, have zero diagonals and satisfy their directed triangle inequalities. In hours and kilometres, define the following quantities by *exact rational arithmetic*:

$$s_v = \frac{6}{5}\,\ell_v + 16h_v, \qquad q_{ij} = \frac{24}{25}d_{ij} + 16\tau_{ij}, \qquad w_{ij} = q_{ij} - 16\tau_{ij} = \frac{24}{25}d_{ij}.$$

Services have fixed endpoints and times; s_v is their prescribed total. They are indivisible non-charging intervals with nonnegative draw throughout. Every bus starts full at the selected depot, serves at least one mandatory service and returns full by the common deadline. Retain the same battery/fleet limits and charge-only, unit-efficiency shared charging resources. There is no taper, switching time or minimum session. Only the selected depot admits charging and zero-draw parking; all other stationary waiting consumes exactly 16 per hour. A second source-flagged depot does not acquire free parking or charging. Post-last-service charging opens at absolute time zero with the same deadline and access policy for every visit label. All operating costs other than grid supply are unchanged by the replacements below; in particular, monetary movement cost is zero.

The broader set $\mathcal{W}$ permits arbitrary *finite* primitive-arc walks and continuous waits between the same fixed services, with no extra task, regeneration, charging location or time-dependent arc attributes. The canonical set $\mathcal{C}$ contains each time-feasible direct and one-via-selected-depot representative: immediate inbound travel, latest outbound departure, the entire intervening depot window, latest full pullout and earliest final return. Its coefficients use this same exact law, so $\mathcal{C} \subseteq \mathcal{W}$.

Completeness includes zero-length depot-endpoint arcs and their access interpretation. A gap starting or ending at the selected depot is a depot-visit case, not an off-depot stationary wait. A constructor that rejects a needed endpoint or a wait at another source-flagged depot must either establish that it never occurs here or supply the appropriate canonical representative; table triangles alone do not prove inclusion or mode completeness. These requirements are premises of the following statement, not consequences of a coefficient-name or mode-count match.

Proposition 20 (Canonical value preservation under the exact law)**.** *Under these assumptions, every feasible schedule in $\mathcal{W}$ has a schedule in $\mathcal{C}$ with the same fleet and service assignment/order and componentwise no greater grid purchases in every market interval. For any coordinatewise nondecreasing supply objective F, physical and complete-plan convexified infimum values therefore satisfy*

$$D^{\mathrm{rat}}_{\mathcal{W}} = D^{\mathrm{rat}}_{\mathcal{C}}, \qquad \mathrm{CH}^{\mathrm{rat}}_{\mathcal{W}} = \mathrm{CH}^{\mathrm{rat}}_{\mathcal{C}}.$$

These are value equalities, not equality of attainable load sets. For nonnegative posted prices the private-score infimum also agrees, but all response ties need not be canonical. No negative-price conclusion is made.

Proof. Both q and w satisfy directed triangles, by their nonnegative linear dependence on the assumed metric tables. Consider a gap of duration T between fixed services that never visits the selected depot. Its total consumption is $16T + \sum w_{ij}$, including its off-depot waits. Repeated triangles make this at least $16T + w_{ab}$ for its endpoints a, b. Direct travel and the remaining off-depot wait attain that value and fit the gap by the time triangle. There is no charging, and consumption is nonnegative: a no-smaller final SOC suffices for all intermediate lower inventory bounds, while no charging can violate capacity.

For a gap visiting the selected depot, replace the prefix to its first arrival by immediate direct travel, and its suffix from its last departure by latest direct travel. Time triangles enlarge the depot window; q triangles and nonnegative waiting weakly lower prefix and suffix energy. Delete all intermediate excursions and remain parked. Keep old charging at its original clocks, curtailing only when the larger SOC would exceed capacity. Unit efficiency, zero depot draw and the absence of minimum sessions make this feasible. The new SOC dominates the old at its charging times because the removed excursions consumed nonnegative energy. The no-more-costly suffix then preserves feasibility of the next service. Each purchase and instantaneous rate only decreases, so shared connector and market-period constraints are preserved.

Latest direct pullout from a full initial battery eliminates unnecessary pre-service consumption. Earliest final return contains the former terminal charging clocks; the common opening-zero policy excludes a visit-label access loophole. Keeping those clocks with capacity curtailment restores full terminal inventory because the original schedule ended full. Zero-diagonal arcs and finite repeated cycles cause no exception; the same argument deletes them without removing an extra task or access right, which the assumptions explicitly exclude. Apply the replacements successively along each duty, preserving its next service's SOC. Services and fleet cost do not change.

Canonical inclusion gives one infimum inequality. The componentwise dominance and monotonicity of F give its reverse. Apply the replacement to every component of a finite complete-plan mixture to obtain the hull inequality: its mean load decreases componentwise and its mean operating cost is unchanged. This proves the stated equalities. The nonnegative posted-price assertion uses the same argument with a linear supply term. □

A fresh certificate for the public-derived target. The original binary supply objective in (17) may be retained; it need not be rounded again when consumption changes. A new target certificate must nevertheless revalidate every lower-bound row and fleet exclusion and admit each proposed upper component under the exact law. We performed those checks on the retained 375-node partition, using its fixed multiplier/tangent proposals and two predetermined route sets. Each route set received one exact charging reconstruction and one independent physical replay under the new law. Their fixed historical mixture weight was retained without optimizing it again. Direct mixed-load evaluation and an independent quadratic-variance identity give

$$\mathrm{CH}^{\mathrm{rat}}_{\mathcal{C}} \le 441.142238 < 441.532912 \le D^{\mathrm{rat}}_{\mathcal{C}} \le 441.532913, \qquad D^{\mathrm{rat}}_{\mathcal{C}} - \mathrm{CH}^{\mathrm{rat}}_{\mathcal{C}} \ge 0.390674. \tag{18}$$

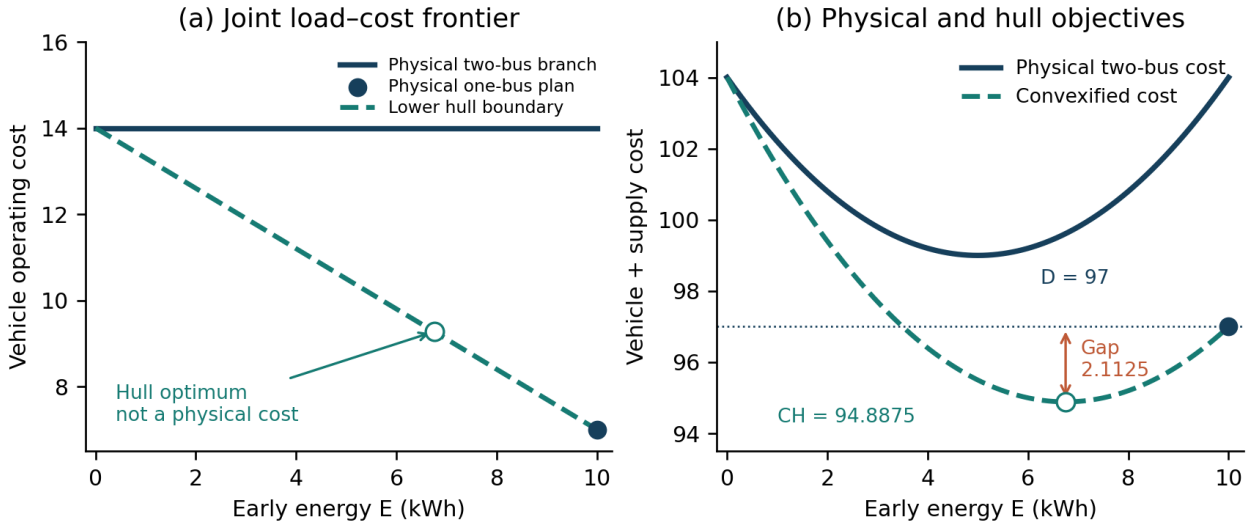


Figure 2: The two-service specialization of Appendix I. Every early load from 0 to 10 is feasible with two buses, but the lower interpolated operating cost is not a physical fleet cost at an interior load. Adding convex supply cost makes this distinction consequential. The curves illustrate proved formulas, not sampled experimental data.

The six-decimal outward numbers coincide with (16), but their exact rational certificates differ. Neither exact optimum is known. The complete canonical time/access qualification and unchanged nonenergy fields establish the representative-coverage premise; the exact law and qualified time/distance tables establish the triangle premises. Proposition 20 therefore transfers the new separation to $\mathcal{W}$. Under the usual convex differentiable supply and attainment assumptions, Proposition 2 then gives the own-marginal-price conclusion. The finite canonical formulation supplies compactness when its charging variables have closed bounded constraints; value preservation does not require the unrestricted finite-walk set itself to be compact. The first target computation retained valid bounds and physical points but failed in a diagnostic-object identity guard before evaluating the mixture. A separately frozen summary-only consumer completed that comparison; it repeated no branch proof, charging reconstruction or physical replay. The failed attempt and its costs remain recorded. This is internal independent verification, not external replication. The existing binary-model conclusion remains exactly as scoped in Section E. Neither result establishes empirical road completeness, energy calibration or operational prevalence.

G Consequences for alternatives and ownership

The frontier distinguishes more charging flexibility from a physically implementable hull optimum. Figure 2 illustrates the two-service specialization in Appendix I. Two services each use 15 kWh; a 20 kWh bus can cover both only by buying 10 kWh between them. Two buses can instead vary early purchases continuously from 0 to 10, but incur an additional vehicle fee. Under the stated full-refill resources, fee 7 and quadratic supply function, the physical value is 97 and the hull value is 94.8875. The energy projection is an interval in both cases; its operating-cost labels make the difference.

Adding alternatives cannot reduce a retained plan's regret at its unchanged own price: the minimum competing bill can only decrease. Nevertheless, a new plan can restore support by imple-

menting a new hull optimum. The exact equipment analysis in Appendix J shows both outcomes. For the stated nominal example, energy-consuming movement additions under unchanged charging access and equipment preserve both planning values. A battery 25 bus at fee 7 instead attains $D = \mathrm{CH} = 92$. Raising its fee to 8 gives $D = 93$ but $\mathrm{CH} = 92.95$: being cheaper than the *old* hull value does not settle the enlarged problem. These claims retain the original services, full inventory accounting and the explicit resource assumptions; they are not universal statements about new routes or battery investments.

Ownership changes the deviation being priced. Appendix K gives a physically realizable replication with all crossmatchings retained, charging resources proportional to fleet size and supply curvature proportional to $1/n$. Its planning gap tends to zero, yet along a specified subsequence one whole owner's absolute own-price regret tends to $351/40$. For separately owned service pairs with *reserved* charging rights, individual regret instead vanishes; per-bus and relative whole-owner regret also vanish. No unilateral-equilibrium claim is made for unreserved shared chargers or an operator anticipating a tariff reset.

The uniform controls matter. For a compact physical load set of diameter R_K, a differentiable μ-strongly convex supply function with Λ-Lipschitz gradient gives the classical explanatory bound

$$\Delta \le r(x; \nabla F(u(x))) \le \Delta + \Lambda R_K \sqrt{2\Delta/\mu}$$

at each physical planner optimum, in one fixed load-space norm (Proposition 31). Thus uniformly bounded diameter and smoothness, and uniformly positive curvature, make that regret vanish with the planning gap. The replication changes these constants with n, so it does not contradict this conclusion. The bound is convex-analysis interpretation, not a new general price-support or learning theorem.

H A separate adequate-resource specialization

We next study a different hardware policy in which the complete assignment set has an exact energy projection. There are $n \ge 1$ mandatory depot-to-depot services A_i occupying $[0, 1]$ and n mandatory services B_j occupying $[2, 3]$. Their consumptions are a_i and b_j, with

$$0 < a_i \le 20, \qquad 0 < b_j \le 20, \qquad a_i + b_j \le 30 \quad \text{for every } i, j. \tag{19}$$

Every A-to-B assignment is allowed, without movement time, energy or cost. At most $2n$ identical buses may be used. Each has capacity 20 kWh, starts full and returns full by hour 4. Its operating fee is strictly positive, $f > 0$.

Early charging in $[1, 2]$ uses n connectors of 10 kW each and an aggregate grid limit of $10n$ kW. Terminal charging in $[3, 4]$ uses n connectors of 30 kW each and an aggregate grid limit of $30n$ kW. Per-bus acceptance matches those connector powers. There is no other charging, reserve, loss, taper or switching time. The grid limits are part of the assumption: connector counts alone would not establish the simultaneous allocation used below.

Since the services in each batch are simultaneous and have positive duration, a bus can cover at most one A and one B. Thus a complete assignment is a partial bipartite matching M. A matched pair uses one bus; unmatched services each use a separate bus. All crossmatchings and all cardinalities $m = 0, \dots, n$ are retained.

Define

$$S = \sum_i a_i + \sum_j b_j, \qquad u_i = \min\{a_i, 10\}, \qquad U = \sum_i u_i, \tag{20}$$

$$r_{ij} = [a_i + b_j - 20]_+, \qquad r_m = \min_{|M|=m} \sum_{(i,j)\in M} r_{ij}, \qquad c_m = f(2n - m), \tag{21}$$

where $r_0 = 0$. The edge cost r_{ij} is the minimum early purchase needed by a bus covering A_i then B_j. The thresholds r_m are minimum-cost cardinality-matching values, not samples from a plan bank.

Theorem 21 (Exact heterogeneous fleet projection)**.** *Under (19) and the stated resources, including $f > 0$, the complete physical projection is exactly*

$$\bigcup_{m=0}^{n} \{(E, S - E, c_m) : r_m \le E \le U\}. \tag{22}$$

The thresholds are nondecreasing and discretely convex:

$$0 \le r_1 - r_0 \le r_2 - r_1 \le \cdots \le r_n - r_{n-1}. \tag{23}$$

Let $m_0 = \max\{m : r_m = 0\}$. The lowest operating-cost boundary of the complete-plan hull is the affine interpolant through (r_m, c_m) for $m = m_0, \ldots, n$, followed by the horizontal segment c_n on $[r_n, U]$.

The physical part of the theorem is the battery and connector argument. A bus that has served A_i can buy any $x_i \le u_i$ early; pairing it with B_j adds the lower bound $x_i \ge r_{ij}$. There are exactly n A buses, so their individual charge intervals can be allocated simultaneously. Terminal replenishment fits by grouping each matched bus alone and each unmatched A/B pair together on one connector. The discrete-convexity step uses standard bipartite min-cost-flow integrality and convex LP value functions. It is not a new matching theorem. The complete proof is in Appendix L.

Use the same scaled supply family as the later replication calculation:

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}, \qquad G(E) = F_n(E, S - E), \qquad E_0 = \frac{S - 20n}{2}. \tag{24}$$

Strict convexity makes the hull minimizing early load E^* unique.

Theorem 22 (A complete support criterion for the two-batch class)**.** *For strictly positive fee $f > 0$ under Theorem 21,*

$$D = \mathrm{CH} \iff E^* \in \{r_m : m = m_0, \ldots, n\} \cup [r_n, U]. \tag{25}$$

The gap is strictly positive exactly when E^* lies strictly inside an adjacent threshold interval (r_{j-1}, r_j), $j > m_0$. If $d_j = r_j - r_{j-1} > 0$, an intermediate threshold r_m minimizes the hull objective exactly when

$$\frac{f}{d_{m+1}} \le G'(r_m) \le \frac{f}{d_m}, \qquad m_0 < m < n. \tag{26}$$

The criterion includes weak support at ties. A threshold is physical, but an interior point between consecutive thresholds interpolates operating costs that no fleet at that early load can attain. Collinear adjacent segments do not erase the intervening physical threshold. If $f = 0$, the

conclusion about strict interiors would fail: the zero-matching branch realizes every early load at zero operating cost, so the gap is always zero. Strict positivity of the fee is therefore substantive.

The values themselves require only matching thresholds and scalar convex minimizations. With $\mathrm{clamp}(v;\ell,u)=\min\{u,\max\{\ell,v\}\}$,

$$D=\min_{0\le m\le n}\{c_m+G(\mathrm{clamp}(E_0;r_m,U))\}. \tag{27}$$

For each sloping hull edge $j>m_0$, evaluate its objective at

$$h_j=\mathrm{clamp}\left(E_0+\frac{5nf}{2d_j};r_{j-1},r_j\right). \tag{28}$$

Compare these values with the minimum on the final horizontal segment. Their minimum is CH. If $m_0=n$, that horizontal segment alone remains and $D=\mathrm{CH}$.

The separate eight-case exact qualification of this hardware policy matched raw SOC enumeration against its threshold formulas in all eight fixed cases, including assignment ties and continuous response sets. All eighteen negative controls and the smoke anchor passed. This is an internal implementation qualification of the stated n-connector policy, not the fixed single-connector model or an operational prevalence estimate.

For the main-text example in (7), the adequate-resource policy also gives $U=15$ and the same cost-labelled projection. Its zero-early endpoint has terminal demands 19, 15 and 15, fitting two 30 kW connectors by serially grouping the last two demands. Its crossed full matching buys 9 early on the 15-energy A bus and replenishes both buses by 20 terminally. These are witnesses for this separate hardware policy. The main text supplies new serial realizations at early power 15 and terminal power 49; the parallel witness timings here are not reused as single-connector schedules. Individual allocation sets can differ even when the joint load–cost projections coincide.

I Two services: capacity, fees and replenishment

The case $n=1$, $a_1=b_1=15$ gives a direct picture of the preceding threshold geometry. Varying its terminal resource also permits a complete connector-capacity characterization. Two mandatory depot-to-depot services consume 15 kWh each: A occupies $[0,1]$ and B occupies $[2,3]$. At most two identical buses are available, each with 20 kWh capacity, zero reserve and full initial inventory. Early charging is available only in $[1,2]$ at 10 kW, both per bus and in total. Terminal charging is available only in $[3,4]$, using one connector of power $K\le 30$ kW. Every used bus must finish full at hour 4. Charging is continuous and lossless, with no taper or switching time. Travel cost, movement consumption and auxiliary load are zero. All prices and costs in this section are synthetic.

There are exactly two service partitions up to vehicle labels. With one bus, inventory after A is 5; it needs at least 10 early kWh to complete B. The early limit therefore fixes load $(10,20)$. With two buses, A's bus buys $x\in[0,10]$ early; terminal purchases are $15-x$ for A's bus and 15 for B's bus. Each bus returns full. Both partitions buy exactly 30 kWh; initial inventory cannot subsidize the horizon.

Let $f\ge 0$ be the cost of a used bus and define

$$F(x,z)=4x+\frac{x^2+z^2}{10},\qquad x,z\ge 0,\qquad g(x)=F(x,30-x)=90-2x+\frac{x^2}{5}. \tag{29}$$

The early/late distinction permits different marginal supply costs despite constant total energy.

Theorem 23 (Nominal failure)**.** *For $K = 30$ and $f = 7$,*

$$D = 97, \qquad \mathrm{CH} = \frac{7591}{80} = 94.8875, \qquad \Delta = \frac{169}{80} > 0.$$

The unique physical planner load is $(10, 20)$ with one bus. Its own-price regret at $p = (6, 4)$ is 13.

The one-bus objective is $7 + g(10) = 97$; the two-bus branch minimizes $14 + g(x)$ at $x = 5$, giving 99. The hull instead mixes one bus at early energy 10 with two buses at early energy 0. Its cheapest operating cost at mean early energy x is $14 - 0.7x$. Thus its objective is

$$14 - 0.7x + g(x) = \frac{7591}{80} + \frac{1}{5}(x - 27/4)^2.$$

The mixture weights are $27/40$ and $13/40$. Both components are physical; the mean operating cost $371/40$ is a convexification. Running those plans on different days would have average physical cost 99.275, not 94.8875.

Figure 2 displays this load–cost geometry in the main text.

At the physical optimum's marginal price $(6, 4)$, its bill plus operating cost is 147. The two-bus zero-early alternative costs 134 at that fixed tariff. This 13-unit incentive changes the entire fleet. By contrast, the hull price $(107/20, 93/20)$ makes the two hull components tie, but is not the marginal supply price of the one-bus physical load. The distinction explains why a hull price and an own-load tariff answer different questions.

I.1 An exact connector and fee characterization

Using the same clamp notation, the next result describes the entire declared parameter domain, including its boundaries.

Theorem 24 (Connector capacity and vehicle fee)**.** *For $0 \le K \le 30$ and $f \ge 0$, the model is infeasible if $K < 20$. If $K = 20$, $D = \mathrm{CH} = f + 90$; both service partitions tie only at $f = 0$. For $K > 20$, put*

$$\ell = 30 - K, \quad d = K - 20, \quad t = \max\{5, \ell\}, \quad h = \mathrm{clamp}\left(5 + \frac{5f}{2d}; \ell, 10\right).$$

Then

$$D = \min\{f + 90,\ 2f + g(t)\}, \tag{30}$$

$$\mathrm{CH} = 2f - \frac{f(h - \ell)}{d} + g(h). \tag{31}$$

For $f > 0$, the gap is strictly positive exactly when

$$\max\left\{0, \frac{2}{5}d(\ell - 5)\right\} < f < 2d. \tag{32}$$

For $f = 0$ the gap is zero. Both physical planner partitions are retained when $f = 90 - g(t)$.

The proof is a physical projection and one-dimensional convex minimization, not a finite parameter scan (Appendix O.2). In particular, with $f = 7$ the gap is zero for $20 \le K \le 47/2$ and positive for $47/2 < K \le 30$. Extra charging capacity can create a price-support failure by enlarging the alternative set; it does not increase the planner's minimum cost or show that installing a connector worsens welfare.

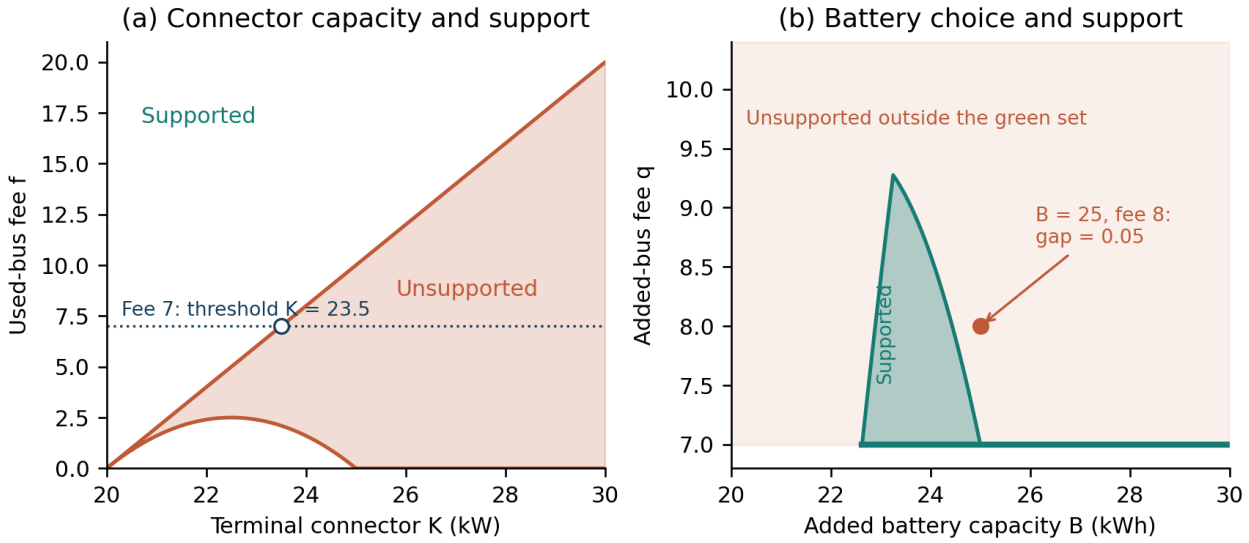


Figure 3: Analytical support regions. Left: the nominal model is feasible for $K \geq 20$; the orange interior has a positive physical–hull gap. At fee 7, support is lost only for $K > 23.5$. Right: green denotes support after adding battery capacity B at fee q while retaining the original buses. The green boundary is included. Better equipment alone is insufficient: $B = 25, q = 8$ still gives gap 0.05. Units and coefficients are synthetic.

Figure 3 contrasts this capacity effect with the equipment extension proved in Section J. The support boundaries are included as weak-support cases; shaded positive-gap regions are strict interiors.

The full price-response value is

$$V(p) = \min\{f + 10p_1 + 20p_2,\ 2f + \ell p_1 + (30 - \ell)p_2,\ 2f + 10p_1 + 20p_2\}. \tag{33}$$

An affine objective attains its minimum on each physical segment at an endpoint. If $p_1 = p_2$, its whole two-bus interval is minimizing within that partition and must be retained whenever its bill also ties globally. This includes the zero-fee boundaries; an endpoint representative alone would omit part of the response set.

I.2 Positive reserve and charging losses

The nominal early limit is binding: adding reserve or losses without changing power can remove the one-bus branch. This is a feasibility issue, not disappearance of the pricing mechanism. A nearby construction uses reserve 1 kWh, efficiency 19/20, early power 12 kW and one terminal connector of 30 kW, keeping the service energies, capacity 20 and fee 7. The one-bus state sequence is

$$20 \longrightarrow 5 \longrightarrow 16 \longrightarrow 1 \longrightarrow 20.$$

It buys 220/19 grid kWh early and 20 terminally. A physical two-bus endpoint buys 30/19 early and 30 terminally; terminal sessions of 9/19 and 10/19 hour fit serially.

Proposition 25 (A nondegenerate neighborhood)**.** *For these modified parameters,*

$$D = \frac{38527}{361}, \qquad \mathrm{CH} = \frac{2987911}{28880}, \qquad \Delta = \frac{94249}{28880} > 0.$$

With terminal power 30 fixed, a positive gap persists in an open neighborhood of the stated reserve, efficiency and early power.

Appendix O.3 states the strict inequalities defining this neighborhood. The result remains conditional on constant efficiency, zero switching time and the absence of charging taper.

J Does failure survive additional alternatives?

Let X be an old set of complete plans, Y a set of additions, and $Z = X \cup Y$. Use the same supply function and compact projected-image assumptions as before, with attained hull-dual prices where needed. Write D_A, CH_A, V_A for the values associated with a set A. Suppose $\Delta_X > 0$. The following distinction is essential: a retained old plan cannot be rescued by more deviations, but a newly admitted plan can become a supported optimum.

Proposition 26 (Expansion and an old dual certificate)**.** *For every old x and unchanged own-load price p,*

$$r_Z(x;p) \geq r_X(x;p) > 0.$$

Erasure of global failure therefore requires a new supported plan and, necessarily, $D_Y \leq \mathrm{CH}_X$. A sufficient condition for survival is $\min\{D_X, D_Y\} > \mathrm{CH}_X$.

Fix an attained old hull-dual optimizer p^*. If every added plan satisfies

$$c(y) + (p^*)^\top e(y) \geq V_X(p^*), \tag{34}$$

then $\mathrm{CH}_Z = \mathrm{CH}_X$. Under this condition, failure is erased exactly when some added plan attains equality in (34) and also satisfies $p^* \in \partial F(e(y))$. If (34) holds with a uniform margin $\varepsilon > 0$, then

$$\Delta_Z \geq \min\{\Delta_X, \varepsilon\} > 0.$$

These are elementary consequences of the classical dual certificate (Appendix P.1). They provide a useful way to ask whether a restricted physical mechanism survives genuinely richer alternatives. With strictly convex supply, supply equality at p^* fixes the old hull-optimal load; fleet equality then fixes its operating cost. Under (34), erasure means physically implementing that projected hull optimum. Without (34), the hull can move, and comparison with its old value is not enough.

J.1 Movement additions under unchanged resources

In the nominal model of Section I, retain all original plans and allow additional movements consuming nonnegative energy and incurring nonnegative extra operating cost. Preserve the two services, at most two buses with 20 kWh batteries, fee 7, full initial and terminal inventories, the early 10 and terminal 30 kW power limits, charging windows and access. There is no energy recovery, inventory transfer or new charging opportunity.

Theorem 27 (Movement-only stability)**.** *Every feasible added plan is componentwise dominated in purchased-energy and operating-cost space by an old physical plan. Consequently*

$$D_Z = 97, \qquad \mathrm{CH}_Z = \frac{7591}{80}, \qquad \Delta_Z = \frac{169}{80}.$$

To see the physical reason, any added plan buys $30+w$ total energy, $w \geq 0$, and at most 10 early kWh. A two-bus plan with early purchase u is dominated by the old two-bus plan $(u, 30-u)$ of

cost 14. A one-bus plan must still obtain at least 10 early kWh to complete both services; the early limit forces $u = 10$, and the old one-bus plan $(10, 20)$ of cost 7 dominates it. The same replacement works inside every hull mixture because F is coordinatewise nondecreasing. The theorem therefore preserves both values, not merely positivity of their difference.

This is an unchanged-resource result. The original no-movement construction already includes both possible service partitions and every continuous charging choice. A new projected point cannot be called an overlooked feasible duty without explaining which physical assumption admits it.

J.2 An explicitly enlarged battery choice

Now change the equipment. Keep buses with 20 kWh batteries and fee 7, and add buses of capacity $B \in [20, 30]$ and fee $q \geq 7$. Two of each type are available, but at most two buses total are used. Every used bus starts and ends at its own full capacity. All service and charging constraints are unchanged.

Put $a = 30 - B$. The new one-bus branch buys $x \in [a, 10]$ early and $30 - x$ terminally, with operating cost q. Every two-bus type assignment permits the old interval $[0, 10]$ and costs at least 14; the old–old assignment therefore dominates mixed and new–new assignments in projected economic comparisons. Their physical records and equal-fee ties remain part of the complete alternative set.

Theorem 28 (Battery and fee conditions for erasure)**.** *For $20 \leq B \leq 30$ and $q \geq 7$, let $a = 30 - B$ and*

$$A(a) = 14 + 2a - \frac{2}{5}a^2, \qquad C(a) = 7 + (10 - a)\left(\frac{2}{5}a - 2\right).$$

The enlarged model has a supported physical plan exactly when

$$(a \leq 5 \text{ and } q = 7) \quad \text{or} \quad (a > 5 \text{ and } 7 \leq q \leq \min\{A(a), C(a)\}). \tag{35}$$

The inequalities include weak support at ties. At fee $q = 7$, this becomes

$$B \geq \frac{55 - \sqrt{95}}{2} \simeq 22.6266. \tag{36}$$

Outside (35), the physical–hull gap is positive.

The new branch's physical minimizer is $t = \max\{5, a\}$. If $a \leq 5$, its marginal prices at $t = 5$ are equal; the new plan minimizes its bill against the cheaper old one-bus plan only at fee 7. If $a > 5$, the two inequalities in (35) compare it respectively with the old zero-early two-bus plan and old one-bus plan. The proof in Appendix P.2 checks the entire continuous opportunity set.

The case $B = 25$ shows why a new plan's physical cost is not the whole story. For $q \geq 7$,

$$D_Z = \min\{97, 85 + q\}, \tag{37}$$

$$\mathrm{CH}_Z = \begin{cases} 85 + q - (q - 7)^2/20, & 7 \leq q \leq 21/2, \\ 7591/80, & q \geq 21/2. \end{cases} \tag{38}$$

At fee 7, the new one-bus load $(5, 25)$ has total cost 92 and is supported at $(5, 5)$. At fee 8, its physical cost 93 is below the *old* hull value 94.8875, but the *new* hull value is 92.95. Its regret is 1 and the gap remains $1/20$. Thus the necessary condition $D_Y \leq \mathrm{CH}_X$ in Proposition 26 is not sufficient.

At fee 12, the old one-bus load $(10, 20)$ and new one-bus load $(5, 25)$ both have physical cost 97. Their own-price regrets are 13 and 5, respectively; $\mathrm{CH}_Z = 7591/80$. Reporting a single selected planner optimum would hide this incentive difference. The new equipment can erase failure, but its presence alone does not do so.

K Aggregation and the owner of the deviation

The homogeneous specialization of Section A provides a separate way to distinguish a planning-value gap from a participant's incentive. Replicate the nominal services to n simultaneous copies of A and n copies of B. Retain the per-bus physical parameters and fee 7. Provide n early connectors of power 10 kW and n terminal connectors of power 30 kW. Supply scales with the market:

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}. \tag{39}$$

This scaling keeps the marginal-price range bounded; it is an assumption of the family, not an asymptotic claim for an unchanged supply curve.

If m buses each cover an A–B pair, the remaining services use $n - m$ A-only and $n - m$ B-only buses. Operating cost is $14n - 7m$, early energy is in $[10m, 10n]$, and terminal energy is $30n - E$. All terminal charges fit: each paired bus takes one connector, while each unpaired A/B pair shares one connector serially. Thus the projected opportunities used below have explicit physical realizations. All A-to-B crossmatchings are admitted in this aggregate model. Identical energies make their projected outcomes depend only on m; fixed pair duties are not being imposed here. In the threshold notation, $r_m = 10m$.

Theorem 29 (Planning gap and whole-owner regret)**.** *The hull value is* $\mathrm{CH}_n = 7591n/80$*. Every physical planner optimum takes an integer* m *nearest to* $27n/40$*, retaining both choices at a tie. Writing* $\delta = m - 27n/40$*,*

$$\Delta_n = \frac{20\delta^2}{n} \le \frac{5}{n}. \tag{40}$$

For one owner controlling all services and buses, the own-price regret is

$$r_n = \begin{cases} m\,40\delta/n, & \delta \ge 0, \\ (n-m)(-40\delta/n), & \delta < 0. \end{cases} \tag{41}$$

In particular, along $n = 40k + 1$*,*

$$\Delta_n = \frac{169}{80n} \longrightarrow 0, \qquad r_n = \frac{351}{40} + \frac{169}{40n} \longrightarrow \frac{351}{40}.$$

For multiples of 40, both quantities are zero.

Approximate convexity of an aggregate does not by itself bound a composite owner's absolute incentive to replace its entire aggregate. This is consistent with aggregation perspectives [21, 11]. In this family regret per bus and regret relative to physical cost vanish; the nonvanishing statement concerns the whole owner's *absolute* fixed-price gain along a declared subsequence.

Now divide ownership among n operators, each responsible for one A/B pair, with a reserved early connector and a reserved terminal connector. Each may use one or two buses and choose its two-bus early charge in $[0, 10]$. Those reserved rights ensure unilateral deviations remain feasible with every other participant's plan held fixed. An aggregate optimum may pair A and B copies across these ownership labels. Before making an individual comparison, implement its same projected outcome by assigning m owners a one-bus plan and the remaining $n - m$ owners a two-bus plan on their own reserved service pairs. Identical service copies and the explicit connector assignment make this possible with unchanged aggregate load, operating cost and marginal price. This is a reimplementation of the economic outcome, not a claim that a cross-matched literal schedule already respects the newly imposed ownership rights. This step is specific to identical services. In

the heterogeneous example of Section A, fixed pairs with energies $(5, 14)$ and $(15, 15)$ require 10 early kWh when both use one bus, whereas the crossed complete assignment requires only 9. The latter outcome cannot be transferred to that fixed-pair institution by relabeling. No individual regret bound for heterogeneous fixed ownership is asserted below.

Corollary 30 (Reserved participants)**.** *For any physical planner optimum reimplemented in the reserved-pair institution as above, each reserved pair's regret at its marginal price is at most* $20/n$*. The regrets sum to* (41)*. Along* $n = 40k + 1$*, each dissatisfied one-bus owner can gain only* $13/n$*.*

The proof is the same per-pair bill comparison (Appendix Q). At $n = 20$ the two tied physical planner optima with $m = 13$ and $m = 14$ have whole-owner regrets 7 and 14, respectively. Neither the whole-owner calculation nor the reserved-pair calculation specifies an institution for unreserved shared chargers or strategic anticipation of a tariff reset.

K.1 Which uniform controls make a small gap informative?

The preceding example does not imply that planning error and incentives are unrelated. The selected price matters in the classical link between lost opportunity cost and duality gaps [12]. The following elementary convex-analysis bound makes the relevant uniform controls explicit; it is an explanatory consequence, not a new general pricing principle.

Proposition 31 (Gap and own-price regret under uniform curvature)**.** *Let the projected complete-plan set* $S = \{(u(x), c(x)) : x \in \mathcal{X}\}$ *be nonempty and compact, and let* K *be its load projection, of diameter* R_K*. Suppose* F *is differentiable,* μ*-strongly convex and has* Λ*-Lipschitz gradient on a relative neighborhood of* $\operatorname{conv} K$*, where* $0 < \mu \le \Lambda$*. Use one fixed Euclidean norm on its affine hull. For any physical planner optimum* x*, at* $p = \nabla F(u(x))$*,*

$$\Delta \le r(x; p) \le \Delta + \Lambda R_K \sqrt{2\Delta/\mu}.$$

If loads lie in a proper affine subspace, gradients and norms can be taken tangentially: normal price components add the same amount to every bill. Economic price admissibility is a separate qualification.

Proof. Let (u^*, c^*) minimize $c + F(u)$ on $\operatorname{conv} S$ and put $q = \nabla F(u^*)$. Hull optimality and linear minimization give $V(q) = c^* + q \cdot u^*$. With the Bregman difference $B_F(u, u^*) = F(u) - F(u^*) - q \cdot (u - u^*)$, this yields the exact identity

$$c(x) + q \cdot u(x) - V(q) = \Delta - B_F(u(x), u^*) \ge 0.$$

Consequently $\|u(x) - u^*\| \le \sqrt{2\Delta/\mu}$. For this fixed plan, regret is the maximum of affine price functions with slopes $u(x) - u(y)$, so it is R_K-Lipschitz in price. Transfer from q to p, use gradient smoothness and discard the nonpositive Bregman term to obtain the upper bound. The lower bound is the tangent inequality from Proposition 2. □

Uniformly bounded R_K, Λ and a uniform positive lower bound on μ therefore make planner regret vanish as $\Delta \to 0$. In our replication family, however, $\mu_n = \Lambda_n = 1/(5n)$ and $R_{K,n} = 10\sqrt{2}\, n$ in two-load coordinates. The displayed upper becomes $\Delta_n + 40|\delta|$. Indeed the fixed-q score gap is zero there, so retaining its Bregman term sharpens this to $40|\delta|$. The exact nonvanishing whole-owner regret is thus consistent with the bound. Arbitrary nonoptimal plans require their own excess $c(x) + F(u(x)) - \mathrm{CH}$ in place of Δ; nonsmooth supply or vanishing curvature needs a different argument.

L Heterogeneous complete-assignment proofs

Proof of Theorem 5. Complete assignments are precisely partial matchings in the declared graph Γ, with every unmatched service retaining a separate bus. For an allowed matching M, a bus serving A_i has inventory $B-a_i$ before early charging. Headroom gives $0 \le x_i \le a_i$; pairing B_j adds $x_i \ge [a_i+b_j-B]_+ \le a_i$. An unmatched B bus is full throughout the early window. Summing these independent intervals gives $[R(M), \sum_i a_i]$. All A buses are present for the entire early window. At $K > 0$ serial charging takes E/K hours, so exactly $E \le K$ is required. When $K = 0$, only zero early energy is possible.

Every bus is present at hour 3. Its terminal demand is $a_i + b_j - x_i$ if paired, $a_i - x_i$ if A-only, or b_j if B-only. These quantities are nonnegative, individually no greater than B, and total $S - E$. At $H > 0$ serial replenishment takes $(S-E)/H$ hours, giving precisely $S - E \le H$. At $H = 0$ no positive terminal demand can be met. Monotone charging preserves inventory bounds and restores each bus full. Thus each matching has exactly interval $[\max\{L, R(M)\}, U]$ when nonempty. Finite minima and a common upper endpoint give the stated cardinality union for $0 \le m \le \nu$. Every such cardinality exists by deleting edges from a maximum allowed matching. Since the empty matching and every singleton option are retained, feasibility is exactly $L \le U$, also when $\nu = 0$ or some vertices are isolated.

Deleting an edge from a minimum size-m matching gives $r_{m-1} \le r_m$. For discrete convexity, use the standard unit-capacity source–A–B–sink network containing only the allowed A–B edges, with their costs r_{ij}. Its fixed-flow LP is integral at integer flow values $m = 0, \dots, \nu$. The LP value is convex in required flow by duality, so $2r_m \le r_{m-1} + r_{m+1}$ for $1 \le m < \nu$. This reasoning uses bipartiteness, not completeness or the sorted-energy shortcut. All thresholds at or below L are dominated in cost by m_0. The first positive retained horizontal increment is $r_{m_0+1} - L$, no greater than $r_{m_0+1} - r_{m_0}$; later increments are the original nondecreasing matching increments. Hence frontier slopes $-f/\text{increment}$ are nondecreasing, followed by zero. Endpoints are physical and their interpolants are hull mixtures. Every physical branch lies above this convex graph, so every mixture does too. The graph is therefore exactly the lower hull boundary. Cardinalities above $m_{\max}$ have empty physical intervals and supply no endpoint. Cardinalities above ν supply no matching at all. If $m_0 = m_{\max}$, including the empty-graph case, only the physical horizontal segment remains. □

Proof of Corollary 6. The convex frontier plus strictly convex continuous G has one minimizing load. At L, at a retained threshold or on the final horizontal segment, a physical plan attains the frontier. Strictly between consecutive retained thresholds, its operating cost is strictly below that of every physical branch because $f > 0$. No physical plan attains the hull minimizer; compactness of the finite union of physical segments gives $D > \mathrm{CH}$. This retains physical thresholds even on a larger collinear hull face. The mathematical value equivalence becomes a price-support equivalence only with the framework's separate qualifications. □

Proof of Corollary 7. Under the corollary's complete-graph assumption, replacing a selected service by an unused smaller energy while keeping its partner cannot increase the coordinatewise nondecreasing edge cost. Thus some optimal size-m matching uses the m smallest energies in each batch. Let $h(t) = [t-B]_+$. For $a \le a'$ and $b \le b'$, monotonicity of the convex increment $h(t+a'-a) - h(t)$ gives

$$h(a+b) + h(a'+b') \ge h(a+b') + h(a'+b).$$

Swapping any two partners ordered in the same increasing direction therefore cannot raise cost. Repeating exchanges yields opposite order and the stated sum. Ties and zero edge costs require no separate assumption. □

Proof of Theorem 21. Every bus serves at most one A and one B, so service assignments are exactly partial bipartite matchings. For a fixed matching M, let x_i be the early purchase on the bus serving A_i. After that service it has inventory $20 - a_i$. Capacity and early acceptance imply $0 \le x_i \le u_i$. If matched to B_j, its inventory after B is $20 - a_i + x_i - b_j$, giving $x_i \ge r_{ij}$. The assumptions imply $r_{ij} \le a_i$ and $r_{ij} \le 10$, hence each interval is nonempty. An unmatched B bus starts full and has no earlier consumption, so it cannot accept early charge. The independent early intervals sum to exactly

$$\left[\sum_{(i,j)\in M} r_{ij}, U\right].$$

There are n A buses and n early connectors, with aggregate grid power $10n$, so every choice of these purchases is realizable.

For terminal charging, a matched bus needs $a_i + b_j - x_i \le 20$ kWh and fits alone on one connector. An unmatched A bus needs $a_i - x_i$; an unmatched B bus needs b_j. Group unmatched A and B buses arbitrarily; each group's sum is at most $a_i + b_j \le 30$ and fits serially on one 30 kW connector during the terminal hour. There are m matched groups and $n - m$ unmatched groups. Aggregate power $30n$ allows all groups simultaneously. Every bus reaches its own full inventory without intermediate overcharge or energy deficit. Full replenishment fixes total purchase S.

All size-m matchings have the same operating cost and upper early endpoint U, and there are finitely many. Their interval union is $[r_m, U]$, proving the exact projection without restricting the matching set.

Deleting one edge from a minimum size-m matching proves $r_{m-1} \le r_m$. To prove discrete convexity, represent the bipartite matching as unit-capacity flow from a source through A and B vertices to a sink, charging edge cost r_{ij}. At integer required flow the LP is integral and has value r_m. Its value as a function of required flow is convex by LP duality, a supremum of affine functions. Therefore $2r_m \le r_{m-1} + r_{m+1}$.

Nonnegative nondecreasing increments can vanish only in an initial block. After m_0, set $d_j = r_j - r_{j-1} > 0$. The interpolant slopes $-f/d_j$ are nondecreasing, and the final slope zero is no smaller. The graph is convex and no higher than any physical horizontal branch; the earlier zero-threshold branches have strictly larger cost because $f > 0$. Each interpolant edge is a convex combination of physical endpoints, and the final horizontal segment is physical. Thus its graph belongs to the hull. Conversely every physical point lies on or above this convex graph, and therefore so does every convex combination. This proves it is exactly the hull's lowest operating-cost boundary. □

Proof of Theorem 22 and the value formulas. The hull boundary is convex; G is strictly convex with derivative $G'(E) = 2(E - E_0)/(5n)$. Their sum has a unique minimizing early load. At a threshold or on the final horizontal segment, a physical plan attains the hull boundary, so $D = \mathrm{CH}$. In an adjacent open threshold interval, the hull operating cost is strictly below the cheapest physical branch c_{j-1}, since $f > 0$. No physical plan attains the unique hull-optimal projected point, and every other load has larger hull objective. The finite union of compact physical segments attains its minimum, proving $D > \mathrm{CH}$. Proposition 2 supplies the support equivalence.

At an intermediate threshold r_m, the objective's one-sided slopes are $G'(r_m) - f/d_m$ and $G'(r_m) - f/d_{m+1}$. Their signs straddle zero exactly under (26), including weak inequalities. Strict convexity on each physical branch gives (27). On a sloping hull edge its derivative is $G'(E) - f/d_j$, whose zero is $E_0 + 5nf/(2d_j)$. Clamping to the edge gives (28); the remaining horizontal segment minimizes $c_n + G$ directly. This covers the $m_0 = n$ case without dividing by a zero threshold increment. □

Details of the heterogeneous example. The stated matrix gives $r_0 = r_1 = 0$ and $r_2 = 9$. Physical minima for $m = 0, 1$ occur at $E = 9/2$ and cost $5841/40$ and $5561/40$, respectively. For $m = 2$, the

minimum is the endpoint $E = 9$, with value $2681/20$. The sloping hull edge objective is

$$\frac{2821}{20} - \frac{151}{90}E + \frac{E^2}{10},$$

whose minimum is $434201/3240$ at $151/18$. Subtracting gives $\Delta = 121/3240$. The endpoint weights stated in the text realize the same load and operating interpolation. At $p = (49/10, 4)$, the price difference is positive, so each branch minimizes its bill at its lowest early endpoint. Their bills plus operating costs are 224, 217 and $2181/10$. Hence the full-set regret is $11/10$, not merely a gain against an arbitrarily selected alternative. All calculations are exact algebra on the displayed ideal model. □

M A classical boundary at three service batches

The two-batch reduction uses pairs of services. A third batch can couple two adjacent matchings through the battery consumption of an entire duty. The following elementary reduction explains this limitation; it is not a new general hardness claim for electric-vehicle scheduling or price support.

Proposition 32 (Three-batch feasibility boundary)**.** *Fleet feasibility is strongly NP-complete even under the following restrictions. Each of three successive batches has n simultaneous, positive-duration services. Service energies and battery capacity B are positive integers encoded in binary, with total service energy nB; time data are rationally encoded. All transitions between successive batches are available and use zero time and energy. At most n buses start full, reserve and nonservice consumption are zero, and no charging or discharging is available before the last batch ends. Every bus must then refill to B using one terminal connector of power one, with the same individual acceptance power and an adequate common terminal window.*

Proof. Use Numerical Three-Dimensional Matching: three labelled multisets of n positive integers must be partitioned into triples, one from each multiset, each summing to B. Strong hardness follows, for example, from Bosboom et al. [8, Section 2, Theorem 2.4]. We may require total sum nB and every number at most B: violations are immediately NO instances and can be mapped to the fixed balanced NO instance $n = 2$, $B = 4$, and multisets $(1, 1), (1, 1), (1, 3)$.

Make the three multisets services in intervals $[0, 1)$, $[1, 2)$ and $[2, 3)$, with their numbers as battery consumptions. Endpoints may all be the same abstract location; its charger remains unavailable before time three. Overlap and the fleet cap force exactly n buses, each serving one service per batch. A duty with energies (a, b, c) is feasible exactly when $a+b+c \le B$, since positive consumption also guarantees all prefix SOC constraints. Summing over buses and using the total nB forces equality for every triple. Thus any feasible fleet gives the required matching.

Conversely, a matching supplies feasible duties that all end empty. Refill the buses serially for B time units each on the one connector during $[3, 3 + nB]$. This restores full terminal inventory without adding an assignment constraint. The construction is polynomial, and on the strongly hard source family B and nB are polynomially bounded. Membership in NP holds for this restricted model: two batch permutations specify the duties, exact triple sums verify them, and the stated terminal serialization completes the certificate. □

The result does not assert hardness of every three-batch variant or of our price-support decision. In particular, allowing intermediate charging changes this reduction. Its point is that complete pairwise compatibility and adequate terminal resources do not remove the energy coupling across three services.

N Proofs for lossless bidirectional charging

The assumptions and result statements remain in Section B.

N.1 Why the projection is exact

Proof of Lemma 8. A paired bus has signed charge in $[a_i + b_j - B, a_i]$: early headroom gives the upper bound, and enough inventory for B gives the lower bound. An A-only bus has $q \in [a_i - B, a_i]$; a B-only bus has $q \in [b_j - B, 0]$. These endpoint conditions keep inventory in $[0, B]$ before and after the remaining service. A monotone action between those inventories is also feasible throughout the early window. Terminal demand is that bus's service consumption minus q, between zero and B.

All upper endpoints are nonnegative and sum to P. The sum of positive lower endpoints is R, while the sum of all lower endpoints is $T = S - N_m B$. Thus total possible negative excursion is $R - T$. For $E \geq R$, set negative coordinates to zero and distribute purchases among the independent nonnegative intervals, whose sum spans $[R, P]$. Gross energy is then E. For $T \leq E < R$, keep mandatory positive coordinates at their lower endpoints and distribute discharge of magnitude $R - E$ among the remaining negative intervals. This is possible exactly because $R - E \leq R - T$ and has gross energy $2R - E$.

Conversely, total positive purchases are at least R, so any allocation with net E has gross energy at least $2R - E$ and at least E. The constructions attain both regimes in (8). Additional charge/discharge cycles cannot reduce gross energy and are unnecessary. □

Proof of Theorem 9. The lemma and early gross budget give $T \leq E \leq P$, $E \leq K$ and $E \geq 2R - K$. All buses are present for the full early window; their monotone actions serialize in at most one hour when $K > 0$. For $K = 0$, the inequalities force $R = 0$, $E = 0$ and zero gross energy, without division. Every bus's terminal refill is nonnegative and at most B. They serialize at H exactly when $S - E \leq H$; at $H = 0$ no positive terminal demand can be met. With a nonempty positive B batch, $E \leq P < S$ makes that zero-terminal-power case infeasible.

For a fixed matching, the exact interval is consequently

$$[\max\{S - H, S - N_m B, 2R(M) - K\}, U].$$

Only $R(M)$ varies at fixed cardinality. Its attained minimum r_m gives the widest interval; all such intervals have common upper endpoint U. Their union is (10). Since r_m is nondecreasing, its threshold minimum is t_0; feasibility is precisely $t_0 \leq U$, attained by a standalone fleet when needed.

Standard allowed-edge bipartite matching convexity gives discretely convex r_m. Interpolate it linearly on $[0, \nu]$. Taking the maximum of the resulting convex function $2r - K$, the affine function $S - (2n - m)B$ and the constant $S - H$ gives a convex function whose integer values are t_m. Each component is nondecreasing, so the thresholds are also nondecreasing. Zero increments can occur only in an initial block. After retaining its cheapest cardinality, the positive threshold increments are nondecreasing. Frontier slopes $-f/(t_m - t_{m-1})$ are therefore nondecreasing, followed by zero. Each endpoint is physical and each edge is its whole-plan mixture; every physical branch lies above this convex graph. This proves the lower-boundary assertion without introducing nonexistent or resource-infeasible cardinalities. □

Proof of Corollary 10. Strict convexity of G gives one minimizing hull load. At a retained threshold or on the final horizontal segment, a physical plan attains the frontier. In an adjacent open threshold interval, $f > 0$ makes its interpolated cost strictly below every physical branch at that load. The compact finite union of physical intervals then has a strictly larger minimum. The classical support

criterion applies only with the stated signed-domain and price qualifications. At fee zero the empty matching physically attains the entire interval at zero operating cost. □

N.2 Identical paid idle buses

Proof of Proposition 11. Consider a matching of size m with k idle buses. Its total fleet is $N = 2n - m + k$, so the cap implies $k \le m$. The box argument of Lemma 8 still applies: idle buses add only negative headroom. The matching's interval is therefore

$$[\max\{S - H, S - NB, 2R - K\}, U],$$

at operating cost fN. Delete any k matching edges and use the resulting $m - k$ matching without idle buses. This fleet has the same size N and fee. Deleting edges cannot increase R, while the inventory term $S - NB$ and upper endpoint U stay fixed. Its feasible interval contains the original interval. Thus every projected point with idle buses has an identical load and cost without them. The reverse inclusion is immediate. □

O Support, physical projection and parameter proofs

O.1 The classical support criterion

Proof of Proposition 2. For any hull point (e, c), linear minimization gives $c + p^\top e \ge V(p)$. If $p \in \partial F(e(x))$, then $F(e) \ge F(e(x)) + p^\top(e - e(x))$. Adding and minimizing over the hull gives

$$\mathrm{CH} \ge V(p) + F(e(x)) - p^\top e(x) = c(x) + F(e(x)) - r(x; p).$$

This proves (4). If $r(x; p) = 0$, equality of physical and hull values follows immediately.

Conversely let $D = \mathrm{CH}$. Compactness and lower semicontinuity give a physical minimizer. The relative-interior qualification gives strong Fenchel duality and an attained maximum $\mathrm{CH} = V(p^*) - F^*(p^*)$. Under the alternative finite-minorant condition, apply the same duality theorem to the finite minorant. It agrees with F on hull loads and its conjugate is at least F^*. The corresponding price gives a dual bound for F at least CH; weak duality forces equality, hence attainment for F as well.

For any physical minimizer x and any attained hull-dual maximizer p^*,

$$\underbrace{c(x) + (p^*)^\top e(x) - V(p^*)}_{\ge 0} + \underbrace{F(e(x)) - (p^*)^\top e(x) + F^*(p^*)}_{\ge 0} = D - \mathrm{CH} = 0.$$

Both terms vanish. Fenchel equality gives $p^* \in \partial F(e(x))$; the first term gives support. For restricted economic prices the converse requires an attained dual price in that subset, as stated. Every dual maximizer satisfies the displayed unrestricted algebraic conclusion; only the admissible maximizers give support under a restricted economic definition. □

O.2 Physical projection and the connector/fee region

Proofs of Theorems 23 and 24. With one bus, the state after A is 5 and the state after B is $x - 10$. Early charging has $x \le 10$, so feasibility forces $x = 10$. Terminal replenishment is 20 and requires $K \ge 20$. With two buses, A's bus has terminal demand $15 - x$, B's bus has demand 15, and $0 \le x \le 10$. Serial charging at power K fits the terminal hour exactly when $30 - x \le K$. These conditions also satisfy each battery's individual headroom and inventory bounds. Thus for $K \ge 20$

the two-bus early interval is $[\ell, 10]$, $\ell = 30 - K$; for $K < 20$ neither partition is feasible. At $K = 20$ both projected loads are $(10, 20)$ and their costs differ only by f.

For $K > 20$, strict convexity of g gives the two-bus minimum at $t = \max\{5, \ell\}$. The one-bus minimum is the fixed point 10, proving (30). Projecting complete plans to early energy and operating cost gives the segment $\{(x, 2f) : \ell \le x \le 10\}$ and the point $(10, f)$. Their convex hull has lowest cost

$$c_{\min}(x) = 2f - \frac{f(x-\ell)}{10-\ell}, \qquad \ell \le x \le 10.$$

The derivative of $c_{\min}(x) + g(x)$ vanishes at $5 + 5f/(2d)$, $d = 10 - \ell$, proving (31).

For $f > 0$, an interior hull minimizer has lower operating cost than every two-bus physical plan at the same early load. Strict convexity excludes the distinct one-bus point from attaining its objective. Therefore the gap is positive exactly when $\ell < h < 10$. Equivalently $\frac{2}{5}d(\ell - 5) < f < 2d$, together with $f > 0$, which gives (32). At an endpoint the physical endpoint attains the hull minimum. For $f = 0$, $h = t$, and the two-bus physical plan attains the hull. Equality of the two physical branch minima gives exactly $f = 90 - g(t)$; within each branch strict convexity fixes its minimizing load. The affine-response formula (33) follows by minimizing over every point of those physical segments.

For $K = 30, f = 7$, the lower hull objective is

$$104 - \frac{27x}{10} + \frac{x^2}{5} = \frac{7591}{80} + \frac{(x - 27/4)^2}{5}.$$

Physical branch values are 97 and 99. The one-bus gradient is $(6, 4)$, and the complete two-bus response at that price is the zero-early endpoint. Bills 147 and 134 give regret 13. The mixture and operating-cost values in the text follow from weight 27/40 on the one-bus point. □

O.3 A reserve/loss neighborhood

Proof of Proposition 25. For reserve $0 \le r \le 5$, constant efficiency $0 < \eta \le 1$, early power P, and terminal power K, define grid-energy quantities

$$T = 30/\eta, \quad \ell = \max\{0, T - K\}, \quad u = \min\{P, 15/\eta\}, \quad h = \max\{\ell, (10 + r)/\eta\}.$$

The two-bus early interval is $[\ell, u]$; the one-bus interval is $[h, u]$ if nonempty. The lower terminal-capacity condition is $T - x \le K$. For the one-bus plan, its state after B is $\eta x - 10$, so reserve imposes $x \ge (10 + r)/\eta$. Charging only A's bus early realizes every interval point; terminal demands fit serially, and the individual bounds follow from the inventory headroom 15 and the stated reserve range.

Write $G_T(x) = F(x, T - x) = G_T(m) + k(x - m)^2$, where $k = 1/5$ and $m = T/2 - 10$. Suppose

$$\ell < m < h < u, \qquad f > 0, \qquad f < 2k(h - \ell)(h - m). \tag{42}$$

On $[\ell, h]$ the hull's lowest operating cost is $2f - f(x-\ell)/(h-\ell)$, obtained from the two-bus endpoint ℓ and one-bus endpoint h; on $[h, u]$ it is f. Put $d = h - \ell$. The hull minimizer $x^* = m + f/(2kd)$ lies in (m, h). Physical and hull values are

$$D = G_T(m) + \min\{f + k(h - m)^2, 2f\},$$
$$\mathrm{CH} = G_T(m) + 2f - \frac{f(m - \ell)}{d} - \frac{f^2}{4kd^2}.$$

Subtracting gives

$$\Delta = \min\left\{k(h-x^*)^2, \frac{f(m-\ell)}{d} + \frac{f^2}{4kd^2}\right\} > 0.$$

At $(r, \eta, P, K, f) = (1, 19/20, 12, 30, 7)$,

$$(\ell, m, h, u, d) = (30/19, 110/19, 220/19, 12, 10).$$

In particular $u - h = 8/19 > 0$ and $2kd(h-m) = 440/19 > 7$. Rational substitution gives the stated values. All inequalities in (42) are strict and the parameter functions are continuous locally. They therefore persist in an open neighborhood of (r, η, P) with $K = 30$ fixed. □

P Expansion and equipment proofs

P.1 The old-plan and dual-certificate statements

Proof of Proposition 26. Set inclusion gives $V_Z(p) = \min\{V_X(p), V_Y(p)\} \le V_X(p)$, so old-plan regret cannot decrease. By Proposition 2, no old plan can become supported. If failure is erased, $D_Z = \mathrm{CH}_Z \le \mathrm{CH}_X < D_X$; thus a new plan attains physical optimum and $D_Y \le \mathrm{CH}_X$. If instead $\min\{D_X, D_Y\} > \mathrm{CH}_X$, then $D_Z > \mathrm{CH}_X \ge \mathrm{CH}_Z$, proving survival.

Under (34), $V_Z(p^*) = V_X(p^*)$, so

$$\mathrm{CH}_Z \ge V_Z(p^*) - F^*(p^*) = \mathrm{CH}_X \ge \mathrm{CH}_Z.$$

For any new physical plan,

$$\begin{aligned} c(y) + F(e(y)) - \mathrm{CH}_X = {} & [c(y) + (p^*)^\top e(y) - V_X(p^*)] \\ & + [F(e(y)) - (p^*)^\top e(y) + F^*(p^*)]. \end{aligned}$$

Both brackets are nonnegative. Equality holds exactly under the fleet and supply equalities stated in the proposition. Compactness ensures that if $D_Z = \mathrm{CH}_X$, some physical plan attains that equality; it cannot be old. With a uniform fleet margin ε, every new physical plan lies at least ε above CH_X, while every old one lies at least Δ_X above it. This gives the quantitative bound. Strict convexity of F makes the minimizer of $F(e) - (p^*)^\top e$ unique on its convex domain whenever attained, proving the projected-point interpretation. □

Proof of Theorem 27. Each added plan buys total energy $30 + w$, $w \ge 0$, because its initial and final battery inventories agree. Its early purchase u is at most 10. For two used buses, the old $(u, 30-u)$ plan of cost 14 is feasible and componentwise no larger in load or cost. For one bus, completion of both services by hour 3 requires $20 + u \ge 30$ even before any additional nonnegative movement consumption. Thus $u = 10$, and the old one-bus plan dominates it. No added feasible plan can reduce the physical objective because F is coordinatewise nondecreasing.

Every hull point is a finite convex combination of physical points. Replace each added component by its old dominator; its mixture has no larger cost or load in any coordinate. Consequently it has no larger convex supply objective. This gives $\mathrm{CH}_X \le \mathrm{CH}_Z$, and inclusion gives the reverse inequality. The same inclusion/dominance argument gives $D_X = D_Z$. This proof does not require the movement timing to be favorable: it concerns every added plan that remains feasible. □

P.2 Battery enlargement and fee thresholds

Proof of Theorem 28 and (37)–(38). The one-new-bus state sequence imposes $x \geq 30 - B = a$ and $x \leq 10$. Every point of $[a, 10]$ is feasible and fully replenished. With two buses, the A bus's headroom after its service is 15 for either type; B's bus starts full before service and cannot buy early energy. Serial terminal charging gives every $x \in [0, 10]$ for every type assignment. Since $q \geq 7$, all mixed/new two-bus projected points are dominated by the old two-bus interval. Therefore

$$D_Z = \min\{97, q + g(t)\}, \qquad t = \max\{5, a\}.$$

Every supported physical plan is a physical planner optimum. No projected old plan can be rescued, and a dominated two-bus addition cannot have zero regret: at equal cost it has an unsupported old counterpart; at larger cost it has a strictly cheaper old counterpart at every price. It suffices to test the new branch's unique physical minimizer t.

If $a \leq 5$, $t = 5$, so $p = (5, 5)$. All plans buy 30 units; comparison with the old one-bus plan is simply $q \leq 7$, hence $q = 7$. If $a > 5$, $t = a$ and $p_1 - p_2 = 2a/5 - 2 > 0$. Within the new branch, the endpoint a minimizes the fixed-price objective. The best old two-bus plan is its zero-early endpoint. Comparison against it gives

$$q \leq 14 - a(p_1 - p_2) = 14 + 2a - \tfrac{2}{5}a^2 = A(a).$$

Comparison against the old one-bus point gives

$$q \leq 7 + (10 - a)(p_1 - p_2) = C(a).$$

These comparisons include all physical alternatives by affine minimization on every branch. They prove (35). For $a > 5$, $C(a) \geq 7$ and $A(a) \geq 7$ exactly when $a \leq (5 + \sqrt{95})/2$ in the relevant domain; also $A(a) - C(a) = 27 - 4a$. At $q = 7$, including $a \leq 5$, this proves (36).

At fee 7 below the battery threshold, the hull's lower boundary joins $(0, 14)$ to $(a, 7)$ and is then flat to 10. The hull objective on the first edge is $14 - 7x/a + g(x)$ and is minimized at $t_h = 5 + 35/(2a) < a$. The physical optimum has early energy a; its gap is $(a - t_h)^2/5$. At and above the threshold the physical point $(\max\{5, a\}, 7)$ attains the hull minimum. Thus the boundary is derived on the whole continuous battery domain, not inferred from a scan.

For $B = 25$, the added endpoint is $(5, q)$. It is below the old lower hull $14 - 0.7x$ exactly when $q < 21/2$. In that regime the new lower boundary joins $(0, 14)$ to $(5, q)$ to $(10, 7)$. The objective decreases along the first edge. On the second it is minimized at $x = 5 + (q - 7)/2$ with operating cost $q - (q-7)^2/10$. Its value is $85 + q - (q-7)^2/20$. At $q = 21/2$ this equals $7591/80$; above it the whole new segment is dominated by the old hull. This proves the formulas. For $q = 8$, substitution gives $\mathrm{CH}_Z = 1859/20$ and $D_Z = 93$. For $q = 12$, both physical minima are 97. At the new load's price $(5, 5)$, the old one-bus bill is cheaper by 5. At the old load's price $(6, 4)$, the old zero-early two-bus bill remains 134; every new one-bus bill is at least $q + 130 \geq 137$. Thus the old plan's regret is 13, including at the fee 12 tie. All quantities use fixed posted prices. □

Q Replication proof

Proof of Theorem 29 and Corollary 30. For m paired buses, the complete projected physical set has early energy $E \in [10m, 10n]$, late energy $30n - E$ and operating cost $14n - 7m$. The connector assignment in the text realizes every such early total: paired buses take 10 each, and unpaired A buses distribute the remainder. Convexification has lower operating boundary $14n - 0.7E$. Substitution in (39) gives the hull minimizer $E = 27n/4$ and value $7591n/80$.

For fixed integer $m \geq n/2$, supply is minimized at $E = 10m$ and total cost is

$$104n - 27m + \frac{20m^2}{n} = \frac{7591n}{80} + \frac{20}{n}(m - 27n/40)^2.$$

For $m \leq n/2$, the best early energy is $5n$ and branch cost is $99n - 7m$, decreasing with m. For odd n, moving from $\lfloor n/2 \rfloor$ to $\lceil n/2 \rceil$ changes the minimized cost by $-7 + 5/n < 0$. For even n the branch formulas meet at $n/2$. Thus an integer nearest to $27n/40$ is optimal, including both at a tie, and $|\delta| \leq 1/2$. This proves (40).

At that optimum the own-load marginal price is

$$p = (4 + 2m/n,\ 6 - 2m/n).$$

The optimal m has $m/n \geq 1/2$, so a cheapest two-bus pair response buys zero early (the whole interval ties when $m/n = 1/2$). The one-bus private objective minus that two-bus objective is $40\delta/n$. If $\delta \geq 0$, precisely the m paired choices are disadvantaged; otherwise the $n - m$ unpaired choices are. This proves (41). With one owner, all choices can be replaced simultaneously. To compare separate ownership, first implement the optimum using m fixed-pair owners with one bus and $n - m$ with two. Each one-bus owner buys 10 early; each two-bus owner buys zero early at the planner optimum, giving the same total load and operating cost as the aggregate optimum even if its literal matching crossed the new ownership labels. Reserved connectors make each unilateral replacement feasible. The identical projected opportunity sets give the same complete-fleet response value, while each owner's gain is at most $20/n$. Summing the disadvantaged owners' gains yields the whole-owner expression.

For $n = 40k + 1$, the optimum is $m = 27k + 1$, so $\delta = 13/40$; substitution yields the displayed subsequence. If 40 divides n, $\delta = 0$. At $n = 20$, the tied $m = 13, 14$ have $\delta = -1/2, 1/2$, giving regrets 7 and 14 respectively. All ties concern physical plans; no fractional bus schedule is being implemented. □

R How the public lower certificate is checked

This appendix explains the algebra behind the public certificate. It is a classical convex lower-bound construction, not a new optimization algorithm. The native solver proposes numbers; exact arithmetic establishes the bound.

A lower bound that tolerates imperfect multipliers. Write a node relaxation as

$$\min_x \ q(x) = c_0 + c^\top x + \tfrac{1}{2} x^\top Q x, \quad Ax \geq b, \quad Ex = d, \quad \ell \leq x \leq u,$$

where $Q \succeq 0$ and every variable has finite bounds. Row signs can be normalized into this form; branch decisions narrow the box or add exact equalities. Choose any rational tangent point z, nonnegative rational multipliers λ and unrestricted rational multipliers ν. Put

$$g = c + Qz, \qquad r = g - A^\top \lambda - E^\top \nu.$$

Convexity and the constraints give the valid node lower bound

$$B = c_0 - \tfrac{1}{2} z^\top Q z + b^\top \lambda + d^\top \nu + \sum_j \min\{r_j \ell_j, r_j u_j\}. \tag{43}$$

Indeed the tangent lower plane is $c_0 - \frac{1}{2} z^\top Q z + g^\top x$. Subtracting $\lambda^\top (Ax - b) + \nu^\top (Ex - d)$ cannot raise it on the feasible set, and minimizing the remaining residual over the finite box gives (43). Neither z nor the proposed multipliers need solve the relaxation exactly. A sign-invalid inequality multiplier cannot be accepted; residual compensation makes approximate stationarity safe without declaring it to be exact stationarity. For a linear objective, take $Q = 0$. All coefficients and comparisons in the accepted bound are rational, including the original stored supply coefficients.

An infeasible node also needs evidence. For $\lambda \geq 0$ and arbitrary ν, every feasible point satisfies

$$(A^\top \lambda + E^\top \nu)^\top x \geq b^\top \lambda + d^\top \nu.$$

If the maximum of the left-hand side over $[\ell, u]$ is strictly below the right-hand side, the node is infeasible. This finite-box contradiction is checked exactly. A native infeasibility status alone does not justify deleting a branch. Failure to obtain a valid contradiction leaves its domain represented by a valid lower bound or an unresolved state.

Coverage is a separate obligation. Branching on a binary movement variable x_j produces the two domains $x_j = 0$ and $x_j = 1$. Both children are retained, and the checker verifies their inherited domains. The minimum valid lower over a complete covering fringe bounds every integer plan in the declared root domain. Search can stop before finding an optimum if every fringe bound already exceeds the fixed comparison target. Conversely, complete domain coverage with weak bounds is not a positive-gap certificate.

The public quadratic tree has the exactly-two-bus root domain. Its separate one-bus obstruction and nonnegative-cost floor for three or more buses are therefore essential to the global physical conclusion. Convexified fleets are not restricted to two buses. A feasible complete-plan mixture supplies an upper on CH; only a global physical lower strictly above that upper establishes $D >$ CH. The tree bound is never used as a hull lower.

The later physical refinement preserves the source-admitted 303-node partition and all 152 frontier identities. Each new local branch begins with its original leaf's inherited decisions. A retained base bound remains valid throughout its descendants; new exact certificates can strengthen that bound but cannot erase a domain. Complete coverage of all original leaves, with 72 new nodes checked, therefore gives a tighter physical lower without repeating the old numerical proof audit. Together with the retained physical witness it yields the outward interval $[441.532912, 441.532913]$. Subtracting the retained up-rounded hull upper 441.142238 from the down-rounded physical lower gives the conservative gap lower 0.390674, not an exact gap or a new hull lower.

What independent verification means here. The checker reconstructs the declared semantic constraints and objective, checks proposed rational bounds and contradictions, verifies every branch domain and completes the fleet-count argument. It does not rerun the native search or assume that a reported solver optimum is exact. Physical upper witnesses and mixture admission have separate checks and source-bound provenance. These are internal independent implementations and reviews; they are not external replication or a claim that the declared physical model is an empirically calibrated transit system.

R.1 Fee-only stability of the certificate

Change only the used-bus fee to $f = 100 + \tau \geq 0$, leaving physics and the supply objective fixed. Let L_2 be the current certified nominal-fee physical lower, U the admitted mixture upper and

$L_3 = 300 + A$ the verified mandatory-energy floor for fleets with at least three buses. The global bound L_2 is valid on the exactly-two-bus subset. Both admitted mixture components use exactly two buses, so their fixed mixture shifts by 2τ. Consequently

$$D_f - \mathrm{CH}_f \;\geq\; \min\{L_2 - U,\; L_3 - U + \tau\}. \tag{44}$$

The second term follows from $Nf + A \geq 3f + A$ for $N \geq 3$ and $f \geq 0$; it would not follow by shifting an arbitrary higher-fleet bound by 3τ. Thus every $f \geq 100$ retains the positive guarantee in (16). A separate exact arithmetic calculation, checked by an independently implemented interval intersection, certifies the open sufficient region $f > 100 + U - L_3$. Its exact rational endpoint is retained in the result packet. This uses a fixed mixture and transferred bounds; it is not the maximal fee interval or a claim that equality at the endpoint restores support. It establishes no robustness to physical-parameter changes. The historical interval calculation used the earlier positive lower bound. Its endpoint depends on U, L_3, not the magnitude of $L_2 - U$ once positive. Replacing L_2 by the refined nominal-fee bound therefore preserves that endpoint and strengthens the guarantee for $f \geq 100$ algebraically; it does not represent a second interval computation.

R.2 Retained evidence history

Table 1 separates native proposals, physical admission and complete bounds. Earlier failures remain part of the record and are not credited with later scientific conclusions. The original search produced a 303-node binary tree using 353 native calls. An independent semantic coverage check verified every node, inherited branch domain and higher-fleet floor, proving a global physical lower strictly above the admitted hull upper. The later bounded refinement kept all 152 frontier identities, added 72 nodes with 102 new native calls, and checked new bounds under their inherited domains. Retained old bounds remained valid on every unresolved subtree. Complete base coverage met the prospectively fixed 10^{-4} physical-width target; the outward interval in (16) has width at most 10^{-6}, without claiming equality of its exact bounds.

Each new physical point was admitted once within the production chain; source-bound qualified witnesses were then transferred without repeating their replay. The separate portable verification subsequently rechecked both complete witnesses in one fresh process. Original tree qualification passed 39 exact controls without a backend call; the distinct refinement interface qualification passed 24 new checks. Search, complete coverage, resource closure and scientific optimality remain separate obligations. The result certifies a separation, not either exact optimum.

An earlier continuous energy-flow relaxation gave the valid hull lower $434.17 \leq \mathrm{CH}_{15}$, not an asserted exact hull value. The initial flat-tariff comparison has a linear objective and therefore identical physical and hull values, with exact outward depot-15 bounds $[404.92, 408.54]$. Depot 16 has an exact flat lower 414.39 and a numerical upper witness: mixed evidence, not a second nonlinear sign certificate. The first sign certificate's downward exact-difference display was 0.008790. The later endpoint bound in (16) supersedes that headline without replacing its retained evidence. The hull optimum and exact gap remain unknown.

Table 1: Public-case evidence stages and their admission boundaries.

Stage	Admitted conclusion and limit
Native integer planning	Floating objective near 441.532912, but the raw candidate failed strict physical admission.
First route recovery	Exact multiplier-sign failure; no physical upper or global price check was admitted by that attempt.
Rational primal and dual	New exact physical witness and fixed-route charging optimum; global fleet optimality remained separate.
Full-graph price query	Exact LP response lower and regret upper; raw native MIP optimum failed strict admission.
Fixed-route price witness	New strict physical deviation and a checked mixture upper; this stage alone did not prove a global gap.
Physical tree and checker	Complete physical lower strictly above that source-bound mixture upper; positive global gap certified.
Partition refinement	All 152 original frontier identities retained; 72 new nodes and 102 new calls narrow the physical interval to width at most 10^{-6}, without a new hull lower.

S Proofs for the common-loss efficient projection

Proof of Lemma 15. In battery coordinates x_v, the boxes are $[a_i + b_j - B, a_i]$ for a paired bus, $[a_i - B, a_i]$ for an A-only bus, and $[b_j - B, 0]$ for a B-only bus. Their positive lower endpoints sum to R, all lower endpoints to T, and upper endpoints to P. Available negative battery exchange therefore totals $R-T$. A one-direction allocation has $C_v = (x_v)_+/\alpha$ and $X_v = \delta(-x_v)_+$. Mandatory purchases give $C \geq R/\alpha$, hence $X = C - E \geq d_R(E)$. Equation (11) proves the terminal lower bound.

For $E \geq R/\alpha$, use no negative exchange and distribute positive battery exchange αE among the independent intervals, whose positive totals fill $[R, P]$. For $E < R/\alpha$, keep mandatory positive coordinates at their lower bounds and distribute negative battery exchange $(R/\alpha - E)/\delta$ over the negative intervals. This is possible exactly when $E \geq R/\alpha - \delta(R - T)$. These constructions attain $d_R(E)$ and span the raw interval $[(1/\alpha - \delta)R + \delta T, P/\alpha]$. For any one-direction allocation the same endpoints are necessary from $C \geq R/\alpha$, $X \leq \delta(R - T)$ and $C \leq P/\alpha$.

The construction's gross grid throughput is $E + 2d_R(E) = \max\{E, 2R/\alpha - E\}$, the minimum possible at that E. The early constraint is therefore exactly $E \leq K$ and $E \geq 2R/\alpha - K$. Also

$$L_R(E) = \max\{S/\alpha - E, S/\alpha + \kappa R/\alpha - (1 + \kappa)E\}.$$

Requiring both pieces to be at most H gives the last two bounds in (14). Each early action is monotone between valid endpoint inventories and can be serialized; each terminal refill is between zero and B/α grid units and all refills serialize when their sum is at most H. Zero-power cases require zero energy without division. This proves necessity and physical attainment. The curve is strictly decreasing in E, so its distinct points at the same fee do not dominate one another. □

Proof of Theorem 16. At a fixed cardinality, every term of $t(R, N)$ and $L_R(E)$ is nondecreasing in R. A matching attaining r_m therefore has a feasible interval containing that of any other matching of the same size, and a terminal load no larger at the same early load and fee. The lemma gives the physical replacement. Standard cardinality-matching convexity applies to r_m. Its linear interpolation, multiplied by nonnegative constants and combined with the affine nondecreasing term $\delta(S - (2n - m)B)$, shows that every threshold component is convex and nondecreasing. Their maximum has these properties, hence its integer samples t_m do as well.

A bus with purchases and exports can have a battery-neutral cycle removed: reduce C_v by ε and X_v by $\alpha\delta\varepsilon$, until one is zero. Battery exchange and terminal load are unchanged, while connector use and early net purchase weakly decrease. Reconstructing a monotone action in the common window preserves inventory feasibility. Removing all such cycles and applying the preceding matching replacement gives a point of Z weakly below every physical point, with unchanged fee. Conversely every point of Z is physical. Removing dominated points from Z therefore yields exactly the physical efficient frontier.

The displayed L_m has at most one kink, giving the finite-segment and convex-hull description. For a coordinatewise nondecreasing supply objective, replacement cannot worsen physical cost. Replacing each component of an arbitrary whole-plan mixture similarly lowers its mean load coordinatewise without changing its mean operating cost, so it cannot worsen hull cost. The reverse inequalities follow from physical inclusion of Z. This proves equality of the reduced values.

For k capped paid idles, the fleet has size $N = 2n - m + k \leq 2n$, hence $k \leq m$. Delete k matching edges; the resulting all-assigned fleet has the same N and fee, no larger R and the same $T = S - NB$. The same replacement argument proves the stated efficient-value redundancy, without asserting equality of all projected loads. □

T A controlled approximation with per-action setup time

Positive setup time changes the exact common-window model. A conservative inner approximation follows from the constructive action bound, provided an explicit occupancy-slack premise holds. The argument is classical convex contraction and Lipschitz control, not a new generic robustness theory or an assertion that switching time is negligible.

Retain the positive directional powers and physical assumptions of Corollary 14. Each nonzero early charge/export action now occupies an additional fixed connector time $\tau \geq 0$, with no energy exchanged during setup. This is a per-action setup, not merely a charge-to-export direction-change penalty. Setups have no extra monetary fee; terminal charging is unchanged. All buses remain available throughout the unit early window. Common affine throughput wear may be included in the intrinsic cost image, but no minimum session, history- dependent degradation or other new constraint is introduced.

Let W_m^0 be the setup-free cardinality polygon, with occupancy $h(C, X) = C/K_c + X/K_d \leq 1$, and let $\Phi_m(C, X) = (E, L, c)$ be its full affine image, including wear when present. Define

$$A_m = N_m + 2\left\lceil \frac{2\alpha K_c}{B} \right\rceil, \qquad W_m^{\mathrm{in}}(\tau) = \{x \in W_m^0 : h(x) \leq 1 - \tau A_m\}.$$

Write $Y_m^0 = \Phi_m(W_m^0)$ and let Y_m^τ be the actual projection of all size-m matchings under the setup law. Ignoring setup gives an outer bound, while the constructive action list gives the sandwich

$$\Phi_m(W_m^{\mathrm{in}}(\tau)) \subseteq Y_m^\tau \subseteq Y_m^0. \tag{45}$$

Indeed its at most A_m early actions take time $h + \tau A_{\mathrm{actual}} \leq 1$. Insert one setup before each nonzero action and retain their serial order. No SOC or terminal quantity changes. The outer inclusion does not assert that minimum-r_m matching remains an exact sufficient statistic once setup is imposed.

Proposition 33 (Slack-anchor approximation)**.** *For each nonempty W_m^0, suppose an anchor $z_m \in W_m^0$ has $h(z_m) \leq 1 - \sigma_m$ with $\sigma_m > 0$. Define*

$$\varepsilon = \max_{m: W_m^0 \neq \varnothing} \frac{\tau A_m}{\sigma_m} \leq 1, \qquad M = \max_{m: W_m^0 \neq \varnothing} J(\Phi_m(z_m)),$$

where $J(E, L, c) = c + F(E, L)$ is convex, lower semicontinuous and finite on the setup-free hull. Let the nonempty outer union be Y^0 and the true setup union be Y^τ. Define physical and hull values as infima over Y and $\operatorname{conv} Y$, respectively. Then

$$0 \leq D_\tau - D_0 \leq \varepsilon(M - D_0), \qquad 0 \leq \mathrm{CH}_\tau - \mathrm{CH}_0 \leq \varepsilon(M - \mathrm{CH}_0).$$

Consequently, with $\Delta_\tau = D_\tau - \mathrm{CH}_\tau$,

$$|\Delta_\tau - \Delta_0| \leq \varepsilon(M - \mathrm{CH}_0). \tag{46}$$

A nominal gap greater than this allowance remains strictly positive.

Proof. For $x \in W_m^0$ use $x_\tau = (1 - \varepsilon)x + \varepsilon z_m$. Convexity preserves every battery and terminal halfspace, while $h(x_\tau) \leq 1 - \varepsilon\sigma_m \leq 1 - \tau A_m$. Thus the same minimizing matching has the physical realization in (45). The affine map gives $y_\tau = (1 - \varepsilon)y + \varepsilon\Phi_m(z_m)$. Convexity bounds its objective by $(1 - \varepsilon)J(y) + \varepsilon M$. Apply this to an outer physical near-minimizer. Setup restrictions only remove plans, proving the physical increment bounds.

For a hull point $y = \sum_i \theta_i y_i$, replace each component and keep its weight. The new mean is $(1-\varepsilon)y + \varepsilon \sum_i \theta_i \Phi_{m_i}(z_{m_i})$. Convexity bounds the anchor mean's objective by M, so the same argument gives the hull increment bounds. The mean is a mixture, not a physical dispatch. Since $D_0 \geq \mathrm{CH}_0$, both nonnegative increments lie in $[0, \varepsilon(M - \mathrm{CH}_0)]$. Their difference proves (46) without a factor two. Comparisons of infima avoid an extra compactness assertion for the true setup set. Empty outer layers are omitted; point and segment layers are allowed if they have the required slack anchor. □

An optional distance bound uses the smaller individual contractions $\lambda_m = \tau A_m / \sigma_m$: every outer point has a setup-feasible point within $\lambda_m \operatorname{diam}(Y_m^0)$ in a fixed norm with declared energy/cost scaling. This remains wear-aware when the lossless load map is noninjective. If J is M_{Lip}-Lipschitz on the outer hull, put $\eta_\tau = \max_m \lambda_m \operatorname{diam}(Y_m^0)$; the physical and hull increments are in $[0, M_{\mathrm{Lip}}\eta_\tau]$, hence the gap changes by at most that same allowance. The primary convex bound needs no Lipschitz premise. Support implications retain the framework's usual price and attainment qualifications.

Why occupancy slack cannot be omitted. Take $n = 1$, $B = 1$, $a = b = 3/4$, unit efficiencies, $K_c = K_d = 1/2$, $H = 2$ and $f = 1$, with the single successor edge and both standalone options. The paired bus needs $Q \geq 1/2$, while $C + X \leq 1/2$ forces $C = 1/2, X = 0$. Its early charge fills the entire window, so every $\tau > 0$ removes this layer. The two-bus layer remains feasible for every τ: use no early action and purchase $3/2$ terminal units. For $F(E, L) = E + L$, full replenishment gives $E + L = 3/2$. Thus physical and hull values jump from $5/2$ to $7/2$ at any positive setup time, even though the model stays feasible. No unconditional continuity claim is possible. A zero-action point itself supplies occupancy slack; the missing paired-layer anchor is the obstruction.

The slack criterion is sufficient, not necessary: zero-action points need no setup, and A_m is conservative. If $\tau A_m > \sigma_m$, clipping ε to one does not prove the required time bound. An empty inner set does not itself prove true infeasibility. For a fixed instance with the stated anchors, $\varepsilon \to 0$ as $\tau \downarrow 0$; no uniform radius across fleet families, unchanged optimizer, efficient-Z formula, minimum action count or runtime follows. This analytical approximation has no numerical qualification and transfers no public-timetable, empirical battery or learning claim.

T.1 Setup time can break minimum-requirement replacement

The setup-free theorem replaces every same-cardinality matching by one with minimum summed early requirement. Positive setup can invalidate that exact replacement even when the assignment graph is complete and both perfect matchings remain feasible. Here setup is charged for each nonzero early charge/export action separately on each bus and occupies the shared connector; it is not merely a penalty for changing charge/export direction.

Proposition 34 (A setup-dependent assignment boundary)**.** *Take $n = 2$, $B = \alpha = \delta = 1$, $a = b = (9/10, 2/5)$, the complete successor graph, early directional powers $K_c = K_d = 1$, terminal power/budget $H = 2$, fee $f = 1$ and setup time $\tau = 3/20$. Retain the common windows, full inventory, zero other consumption and no terminal setup. The uniquely minimum-requirement perfect matching cannot reproduce every feasible load–cost point of the other matching.*

Proof. The two perfect matchings have the following requirements:

Matching	Summed early requirement	Mandatory charging recipients
Diagonal	4/5	One
Cross	3/5	Two, each requiring 3/10

On the diagonal, purchase $C = 4/5$, $X = 0$ on the high bus in one early action. Setup plus charge takes $3/20 + 4/5 = 19/20$. After A its SOC is $1/10$; charging raises it to $9/10$, and its B service leaves zero. The low bus moves from $3/5$ after A to $1/5$ after B without early charging. Terminal purchases 1 and $4/5$ restore full inventory, using $(9/5)/2 = 9/10$ of the terminal window. Thus $(E, L, c) = (4/5, 9/5, 2)$ is physically feasible.

The cross matching requires positive early charge on two different buses. At $C = 4/5, X = 0$ it needs at least $4/5 + 2(3/20) = 11/10$ early time. Splitting actions or lowering power cannot help. Nor can cycling rescue the net-load point: $E = 4/5$ implies $C = 4/5 + X$, increasing gross use when $X > 0$, while both mandatory recipients remain. Setup cannot overlap another action on the shared connector, and terminal charging cannot repair an earlier service inventory violation.

The cross matching is nevertheless feasible at $C = 3/5, X = 0$: charge $3/10$ on each bus in total early time $9/10$, then purchase 2 terminal units at power 2. A partial matching or paid idle bus cannot reproduce $c = 2$, since the two simultaneous services in each batch force at least two buses, and any such extra-bus alternative costs at least 3. □

A monotone linear query also exposes the omission. For $F(E, L) = E + 2L$, full unit-efficiency refill gives $E + L = 13/5$, so total objective is $c + 26/5 - E$. Any early action leaves at most $17/20$ energy-transfer time; the diagonal attains $E = 17/20$ with one high-bus charge. The cross requires two actions and attains at most $E = 7/10$, realized by purchases $3/10$ and $2/5$. Their optimal values are therefore $127/20$ and $13/2$. Every fleet costing at least 3 has value at least $147/20$, so lower-cardinality alternatives do not repair the restricted minimum. The error from retaining only minimum-r_m matchings is $3/20$. Because the objective is linear, the true physical and hull values agree; this is not a positive support-gap example.

The setup-free outer polygons and the slack-anchor inner bounds in Appendix T remain valid. What fails is exact matching replacement: concentration of recharge among buses matters when each recipient/action consumes setup time. This supplies neither a minimal replacement statistic nor a hardness, public-timetable, empirical or learning claim. For charge-only operation, Appendix T.2 gives exact classical fixed-matching and joint-duty baselines that retain this setup information explicitly.

T.2 Classical exact formulations for charge-only setup

Loss of the minimum-r_m summary does not require enumerating all matchings or using a learned method. The common-window charge-only model admits ordinary resource allocation and a compact duty formulation. These are classical baselines, not new optimization algorithms.

Assume common capacity B, zero other draw/reserve, full initial and terminal inventories, common efficiency $\alpha \in (0, 1]$ in both windows, early accepted grid power $K_c > 0$, terminal accepted power/budget $H \geq 0$, and common setup $\tau \geq 0$ assessed separately for each bus's nonzero early action. Each used bus pays $f > 0$. Both windows have unit duration and all used buses are available throughout them. Both windows permit charging only, and allowed transitions have zero time, energy and cost. Terminal actions have no setup. No minimum session, taper or intrawindow access change is allowed. Early sessions on one bus can then be merged without changing endpoint SOC or increasing transfer time or setup count.

T.2.1 An exact interval for a fixed matching

Fix the duties. A bus consuming A_v before the early window and B_v afterward has the true grid-energy box

$$r_v = \frac{(A_v + B_v - B)_+}{\alpha}, \qquad u_v = \frac{A_v}{\alpha}, \qquad r_v \le q_v \le u_v.$$

Absent services have energy zero; require $A_v, B_v \in [0, B]$. Let $R = \sum_v r_v$, let h count the mandatory recipients $r_v > 0$, and put $U_h = \sum_{r_v>0} u_v$. Sort the positive optional capacities as $v_1 \ge \cdots \ge v_p > 0$. Zero-capacity buses remain in the fleet and fee count, but need not enter this scan. If $R > K_c(1-\tau h)$, early charging is infeasible. Otherwise define

$$C_{\max} = \max_{h \le k \le h+p} \min \left\{ U_h + \sum_{j=1}^{k-h} v_j,\ K_c(1 - \tau k) \right\}.$$

The attainable early totals are exactly $[R, C_{\max}]$. A k-recipient schedule cannot exceed either the largest available capacity or its remaining transfer time. Conversely those top optional capacities and the mandatory intervals attain every total above R up to their sum, with arbitrarily small positive optional allocations when needed. If the maximum equals R, use only mandatory recipients. Decreasing charges toward their lower bounds never increases active count, proving there are no holes. Negative later scan terms cannot improve the feasible mandatory baseline. In particular $h = 0$ admits zero actions for every τ, even when all optional capacities vanish.

Write $G = \sum_v (A_v + B_v)/\alpha$ for total grid replacement energy, not the planner value D. Terminal refill gives the exact final interval

$$\max\{R, G - H, 0\} \le C \le \min\{C_{\max}, G\}, \qquad E = C, \quad L = G - C, \quad c = fN,$$

when its endpoints are ordered. Valid individual boxes, not merely aggregate clipping, ensure physical inventories. For $K_c = 0$ separately, only $C = 0$ is possible, requiring all $r_v = 0$ and $G \le H$. The scan uses mandatory count and ordered capacities as well as R; it is not an exact joint-matching reduction.

T.2.2 A compact duty MILP

Use each allowed pair (i, j) and each standalone $(i, *)$, $(*, j)$ as a candidate duty d, at most $n^2 + 2n$ in total. Let A_d, B_d be its energies before/after the early window, with zero for an absent service, and $s_d = A_d + B_d$. Variables are duty activation $y_d \in \{0, 1\}$, setup permission $z_d \in \{0, 1\}$, early purchase $q_d \ge 0$ and terminal purchase $t_d \ge 0$. Impose exact service cover

$$y_{i,*} + \sum_{j:(i,j)\in\Gamma} y_{ij} = 1, \qquad y_{*,j} + \sum_{i:(i,j)\in\Gamma} y_{ij} = 1,$$

and, for every duty,

$$\begin{aligned} z_d &\le y_d, & \alpha q_d &\le A_d z_d, & \alpha q_d &\ge (s_d - B) y_d, \\ \alpha(q_d + t_d) &= s_d y_d, & 0 &\le \alpha t_d \le B y_d. \end{aligned}$$

The shared constraints are

$$\sum_d q_d / K_c + \tau \sum_d z_d \le 1, \qquad \sum_d t_d \le H.$$

An additional fleet limit may bound $\sum_d y_d$; cover alone gives $n \le \sum_d y_d \le 2n$. Define $E = \sum q_d$, $L = \sum t_d$ and $c = f \sum y_d$, optionally adding common affine energy wear.

Every physical bus induces one duty. Merging its charge-only early sessions yields these inequalities and sets $z_d = 1$ for positive q_d. Conversely, serialize each positive q_d with one setup at power K_c. The capacity and reserve inequalities keep SOC between zero and B before and after B; nonnegative service consumption preserves prefixes. Serialize terminal charges using the shared budget; the equality restores every used battery exactly full. A permitted $z_d = 1, q_d = 0$ only overcounts setup time: omit that action, preserving projected feasibility. Thus the formulation is exact for aggregate energy/cost decisions, not for rewarding the literal count of nonzero sessions.

Any affine aggregate query is an MILP; convex quadratic grid cost gives a convex MIQP. Polynomial formulation size does not mean polynomial solution time, nor is its continuous relaxation automatically the complete-plan hull. Exact sets here exclude counted idle buses; with positive fee they cannot improve $c + F(E, L)$ in this charge-only, full-start/end model, because they exchange no energy and only add cost. The result does not extend the session-merging argument to V2G or state/session-dependent degradation. It supplies no hardness, numerical advantage, public-data or ML claim.

U Convex dispatch can have negative marginal prices

The monotonicity premise in Theorem 16 is substantive when supply cost is obtained by optimizing a network dispatch. Transmission congestion can yield negative nodal marginal prices even with nonnegative generation offers; this is an established effect [5, Section V-B, Figure 5]. The following independently checked symbolic example makes the distinction explicit with strictly increasing, strictly convex generator costs. It is a scope illustration, not a new negative-price result or empirical network.

Consider a lossless three-bus DC network with unit susceptance on each of lines $12, 13, 23$, oriented flows $f_{ij} = \theta_i - \theta_j$, and $\theta_3 = 0$. Symmetric line limits are $5, 1/2, 5$, respectively. Background demand is $(4, 0, 1)$ and an additional depot withdrawal $x \in [0, 1]$ occurs at bus 3. Generators at buses 1 and 2 satisfy $0 \le g_1, g_2 \le 10$, with costs

$$C_1(g_1) = 3g_1 + g_1^2/100, \qquad C_2(g_2) = g_2 + g_2^2/100.$$

All quantities use consistent power and per-period cost units; a fixed period converts them to energies. Nodal balance is generation minus demand equal to outgoing flow. Let $J(x)$ be the minimum total generation cost under these constraints.

Bus-3 balance and total balance give $\theta_1 + \theta_2 = 1 + x$ and $g_2 = 5 + x - g_1$. Bus-1 balance then implies

$$\theta_1 = (g_1 - 3 + x)/3, \qquad g_1 \ge 3/2 - x,$$

where the inequality follows from the lower limit on line 13. The point

$$g_1 = 3/2 - x, \quad g_2 = 7/2 + 2x, \quad \theta_1 = -1/2, \quad \theta_2 = 3/2 + x$$

attains that lower bound and is feasible throughout the stated interval: its flows are $(-2 - x, -1/2, 3/2 + x)$, and all generator limits hold. After eliminating g_2, the objective derivative with respect to g_1 is

$$2 + \frac{2g_1 - 5 - x}{50} \ge 2 - \frac{6}{50} > 0.$$

Thus every feasible point has cost at least that of the displayed point; this establishes global optimality, not merely stationarity at a proposed active set. Direct expansion yields

$$J(x) - J(0) = -\frac{89}{100}x + \frac{x^2}{20}.$$

This incremental cost is strictly convex and strictly decreasing on $[0, 1]$. More withdrawal at bus 3 relieves the congested reverse flow toward bus 1, reducing output at the expensive generator at bus 1 while increasing cheaper generation at bus 2. The redispatch saving exceeds the cost of the extra supply. Both generators' own cost derivatives remain positive.

Consequently convexity of the optimized supply value and positive generation marginal costs do not ensure monotonicity in nodal load. A network application of the efficient-frontier reduction must establish the required load-domain monotonicity, or retain the alternatives that coordinatewise dominance would discard. The example does not invalidate the theorem under its hypotheses, change the public synthetic supply objective, or establish AC feasibility, prevalence or a learning advantage.

V Proof of the common-transition extension

Proof of Corollary 18. For a fixed matching, paired, A-only and B-only early battery-exchange boxes are, respectively,

$$[a_i + b_j + d - B, a_i], \qquad [a_i - B, a_i], \qquad [b_j - B, 0].$$

The edge condition makes the first box nonempty. Capacity after A gives the upper endpoints; nonnegative SOC after the transition and B gives the lower endpoints. Nonnegative consumption during the post-early portion then guarantees all its intermediate SOC constraints. The upper endpoints sum to P, all lower endpoints to $T = S_m - N_m B$, and their positive parts to $R = \sum_{(i,j)\in M}(a_i + b_j + d - B)_+$.

The allocation argument of Lemma 15 therefore gives minimum export $D_R(E) = (R/\alpha - E)_+$ and terminal load $S_m/\alpha - E + \kappa D_R(E)$. The available negative battery exchange, early gross budget and terminal budget give the four lower bounds in t_m with R in place of r_m; early capacity and gross budget give $E \le \min(P/\alpha, K)$. Every point in that interval is attained by distributing positive or negative exchange over the boxes, executing early exchanges serially, performing the noncharging task and B service, and refilling serially at the terminal. Extra cycles weakly increase terminal load and resource use, so monotone value objectives permit their removal.

At fixed m, total energy S_m and fleet N_m are constant across matchings. Both the terminal curve and each threshold term are nondecreasing in R, since $\alpha^{-1} - \delta \ge 0$ and $\kappa \ge 0$. A matching attaining r_m consequently dominates all other size-m curves. Componentwise replacement also proves the convexified-value claim. Classical cardinality-matching convexity makes r_m discrete convex, while S_m and $S_m - N_m B$ are affine. Each threshold term is discrete convex and so is their maximum: extend each term by its convex linear interpolant, take the convex maximum, and restrict to integers.

For completeness, suppose k paid idle buses are admitted with total $N = 2n - m + k \le 2n$, all available early. Their boxes are $[-B, 0]$. Delete any $k \le m$ matching edges and replace each deleted pair plus one idle bus by its two singleton services. Fleet and fee stay unchanged, P stays unchanged, total energy drops by kd, and R weakly decreases. Every threshold term weakly decreases and the terminal curve weakly decreases at the same E. The no-idle representative therefore dominates under monotone economics. This proves efficient-value dominance, not equality of feasible load sets; it makes no claim about uncapped idle fleets. □

W Many efficient branches with varying transition energy

This analytical construction sharpens the preceding transition-energy boundary: at one fixed fleet size, varying compulsory transition energy can produce many distinct efficient physical pieces. It is an elementary independent-choice construction with a feasible charging realization, not a new generic Pareto-complexity or scheduling-hardness claim. No numerical implementation qualification is asserted.

Proposition 35 (Efficient pieces at fixed cardinality)**.** *For every $k \geq 1$, there is a two-batch charge-only instance with $2k$ services in each batch, fleet cap $2k$, and 2^k distinct efficient physical branches at that same fleet size. Every branch has a corner that is the unique projected load/cost optimum for some coordinatewise strictly increasing convex supply objective. The input has polynomial binary encoding length in k; numerical magnitudes grow with k.*

Proof. Let $w_i = 2^i$ for $i = 0, \ldots, k-1$, $W = \sum_i w_i$, and choose common battery capacity $B \geq 20 \max_i w_i$, for example equality. Block i has two A services and two B services, with energies $(B/2 + 3w_i, w_i)$ in each batch. The allowed graph is the disjoint union of the k complete 2×2 blocks. Within a block, identity edges have transition energy 0 and cross edges have transition energy $6w_i$. No interblock edge or alternative transition is admitted. A transition is a compulsory task, not an excursion that may be omitted.

All buses start and end full, with zero reserve, unit efficiency, no V2G and no other nonservice draw. Use successive windows A: $[0, 1]$, early charging: $[1, 2]$, noncharging transition tasks: $[2, 3]$, B: $[3, 4]$, terminal charging: $[4, 5]$. Tasks have nonnegative consumption throughout their positive duration. Access is the ideal common-window access assumed in the two-batch model; no road-metric realization is claimed. There is one early connector of power/budget $K = kB/2 + 4W$ and one terminal connector of power/budget $H = kB + 20W$, with per-bus acceptance at least these powers. A common fee $f \geq 0$ is paid per counted bus.

Each batch has $2k$ simultaneous services and the fleet cap is $2k$. Thus exactly $2k$ buses are used, every bus serves one service from each batch, and no singleton or idle bus can compete. Each block chooses its identity or cross perfect matching. Identity requires early purchase $6w_i$ and consumes $B + 8w_i$ in total. Cross requires zero early purchase and consumes $B + 20w_i$. Indeed each cross duty consumes $B/2 + 10w_i \leq B$ before terminal charging. The capacity assumption also ensures each post-early portion fits the battery; it is stronger than merely checking that condition.

For the set I of identity blocks put

$$R = 6 \sum_{i \in I} w_i, \qquad S_{\max} = kB + 20W.$$

Total consumption is $S_{\max} - 2R$, while the sum of early headrooms is K, independent of matching. The per-bus early purchase intervals sum to $[R, K]$. Thus the *complete* fixed-matching load/cost branch is

$$(E, L, c) = (E, S_{\max} - 2R - E, 2kf), \qquad R \leq E \leq K.$$

Every point is physically attained: distribute early purchases within those intervals and serialize them on the common early connector, perform the noncharging tasks and services, then serialize the full terminal refills. Nonnegative consumption and the endpoint battery bounds imply all intermediate SOC bounds. The interval is nonempty since $6w_i \leq B/2 + 4w_i$. Terminal purchases are nonnegative and at most H. Conversely every schedule has one of these perfect matchings and exactly these energy balances, so there are no additional physical branches.

Binary subset sums give 2^k distinct values $R = 6t$, $0 \leq t \leq W$. Their corners are $z_R = (R, S_{\max} - 3R, 2kf)$. For consecutive values $R < R_{\text{next}}$, the portion $R \leq E < R_{\text{next}}$ is efficient:

larger-requirement branches are unavailable at that early load or a smaller one, while smaller-requirement branches have a larger terminal load there. Reducing E on the same branch increases its terminal load. The final branch is efficient through $E = K$. At R_{next} the new corner dominates the preceding branch endpoint. Hence these are actual efficient pieces, not merely differently labelled matchings.

Fix a corner R_* and set $\varepsilon = 1/(2\max\{K, 1\})$. On the nonnegative load domain the supply objective

$$F_{R_*}(E, L) = 3E + L + \varepsilon(E - R_*)^2$$

is convex and strictly increasing in each coordinate: $\partial_E F \geq 3 - 2\varepsilon R_* \geq 2$ and $\partial_L F = 1$. On a branch indexed by R,

$$F_{R_*}(E, L) = S_{\max} + 2(E - R) + \varepsilon(E - R_*)^2 \geq S_{\max}.$$

Equality requires $E = R = R_*$, so that corner is the unique projected optimum. The constant fleet cost does not affect this conclusion, including when $f = 0$. Different charging allocations with the same loads can still tie; uniqueness of a literal schedule is not claimed. □

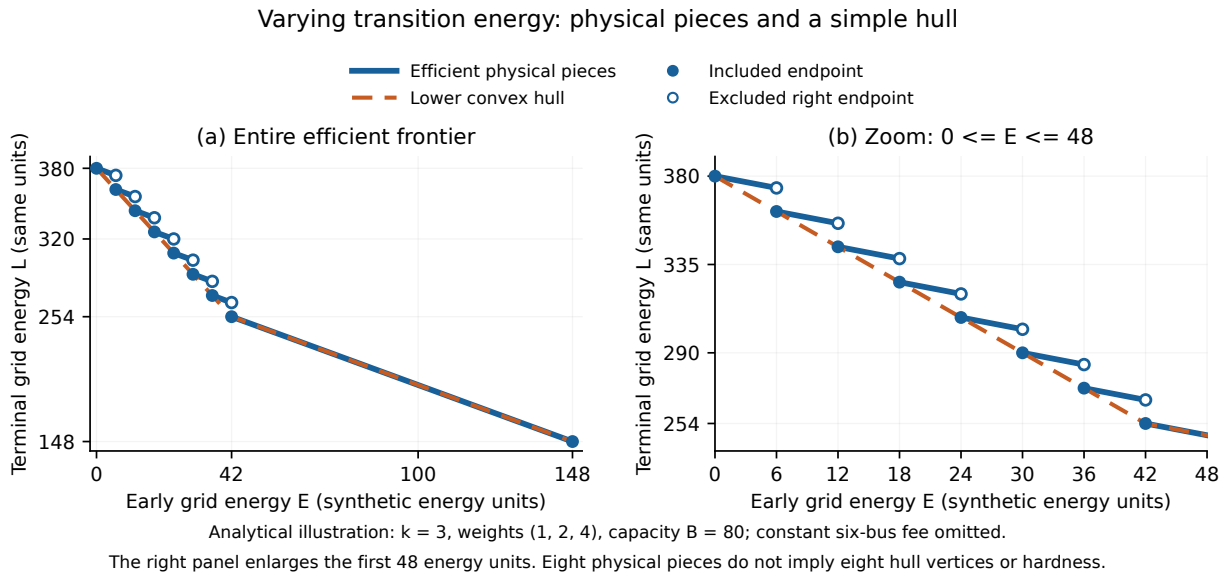


Figure 4: The analytical construction with $k = 3$, $B = 80$, $K = 148$ and $S_{\max} = 380$. Its eight efficient physical pieces have collinear corners. Open circles exclude the dominated right endpoint of each nonfinal piece. The simple lower hull illustrates why this branch count is not a hull-vertex or optimization-hardness result. The common six-bus fee is omitted.

What the branch count does not imply. All corners lie on $L = S_{\max} - 3E$ and share the supporting price $(3, 1)$. They therefore do not give 2^k hull vertices or distinct linear-price regions: the hull can replace their collinear corners by one edge. The quadratic term selects a particular projected corner, rather than making each corner uniquely exposed by a different linear tariff. Weights, capacity and connector powers grow with this family. Explicit binary input length is polynomial in k, so the explicit physical-piece count has no polynomial bound in that length; this is not a $2^{\Omega(\text{input bits})}$ lower bound, a strong-hardness theorem or an optimization-hardness argument.

Indeed $R = 6t$, $0 \le t \le W$ is a compact symbolic description. The result neither excludes symbolic compression nor says this particular family is difficult to optimize. Its narrow lesson is that one minimum-requirement matching per cardinality can discard efficient and uniquely convex-optimal projected points when total energy becomes assignment-dependent.

Corollary 36 (A support gap with all matching alternatives present)**.** *For the complete family in Proposition 35, take the smooth, coordinatewise strictly increasing convex supply cost*

$$F(E, L) = 3E + L + \frac{(E-3)^2}{2K}.$$

For every $k \ge 1$ and $f \ge 0$, the whole physical and complete-plan convexified planning values under the stated fleet cap are

$$\mathrm{CH} = S_{\max} + 2kf, \qquad D = \mathrm{CH} + \frac{9}{2K} > \mathrm{CH}.$$

The hull optimum is unique in projected coordinates. Exactly the physical corners $R = 0$ and $R = 6$ minimize physical planning cost. Under Proposition 2, no physical plan is supported at its own marginal prices in this declared opportunity set.

Proof. Here $K \ge 14$, and both corners exist for every $k \ge 1$. The Hessian of F is $\mathrm{diag}(1/K, 0)$, and its derivatives on the nonnegative load domain satisfy $\partial_E F \ge 3 - 3/K > 0$ and $\partial_L F = 1$. Every physical branch obeys the affine lower bound

$$3E + L = S_{\max} + 2(E - R) \ge S_{\max}.$$

The bound holds on its complete convex hull as well, giving total cost at least $2kf + S_{\max} + (E - 3)^2/(2K) \ge 2kf + S_{\max}$. The equal mixture of the physical corners $(0, S_{\max})$ and $(6, S_{\max} - 18)$ has load $(3, S_{\max} - 9)$ and attains equality. Equality forces that same E and affine bound, so the projected hull optimum is unique. This argument covers every branch and its convex mixtures, not just the two used by the upper witness.

On a physical branch total cost equals

$$2kf + S_{\max} + 2(E - R) + \frac{(E-3)^2}{2K}.$$

Its derivative in E is at least $2 - 3/K > 0$, so it is minimized at $E = R$. Among $R = 6t$, $0 \le t \le W$, exactly 0 and 6 are nearest to 3, at distance 3. Their objective excess is therefore $9/(2K)$, establishing the stated physical value and its two minimizing projected corners. All plans have the same fee, including when $f = 0$. The strict gap and the classical support criterion give the final assertion. □

This is a whole-opportunity-set example: its 2^k efficient matching alternatives are all admitted, and the fleet cap excludes singleton or extra-bus escape routes. Having many additional feasible alternatives therefore does not by itself guarantee restoration of own-marginal-price support. It is not a claim that every expansion preserves a gap. The exact gap $9/(2K)$ shrinks as K grows; there is no uniform lower bound, prevalence, operational-routing or hardness conclusion. The common supporting price $(3, 1)$ at the hull optimum and the simple hull remain consistent with the earlier collinearity and compression caveats.

X Support failure on a complete assignment graph

The disjoint blocks in Proposition 35 make its independent choices transparent, but disconnected assignment compatibility is not necessary for its support obstruction. The following extension admits every cross-batch assignment, including every perfect matching. Its edge consumptions remain compulsory abstract tasks, not metric road movements or optional routes.

Proposition 37 (A complete-graph extension)**.** *Retain k, w_i, W, B, K, the $2k$ simultaneous services in each batch, and the charge-only common-window model of Proposition 35, including $B \geq 20w_{\max}$ and fleet cap $2k$. Replace the allowed graph by the complete bipartite graph. Keep the original within-block transition energies, and on each interblock edge to B service j set*

$$d_{ij} = B - b_j.$$

Increase the shared terminal power/budget to $H = 2kB$, retaining the unit terminal window and adequate per-bus acceptance. Every perfect matching is physically feasible. For $f \geq 0$ and $F(E, L) = 3E + L + (E-3)^2/(2K)$, the entire complete-graph problem has

$$\mathrm{CH} = S_{\max} + 2kf, \qquad D = \mathrm{CH} + \frac{9}{2K} > \mathrm{CH}.$$

All 2^k original corners remain uniquely optimal in projected load/cost coordinates for their respective monotone convex selector objectives in Proposition 35. This does not assert that the full efficient frontier is unchanged.

Proof. The fleet cap and simultaneous batches force exactly $2k$ buses, each serving one A and one B service. No singleton, partial matching or idle bus can avoid this restriction. All edges perform their compulsory nonnegative task during the same positive-duration noncharging interval between early charging and B. Access and all other draw remain as in the original common-window construction.

For an edge put $e_{ij} = a_i + b_j + d_{ij}$ and $r_{ij} = (e_{ij} - B)_+$. Its early purchase interval is $[r_{ij}, a_i]$ whenever $b_j + d_{ij} \leq B$. This condition holds for all within-block edges by $B \geq 20w_{\max}$, and with equality for every interblock edge. In the latter case $e_{ij} = a_i + B$ and $r_{ij} = a_i$: the bus refills completely in the early window.

For any perfect matching M, these intervals sum to $[R_M, K]$, where $R_M = \sum_M r_{ij}$ and $S_M = \sum_M e_{ij}$. The complete matching branch is $L = S_M - E$, $R_M \leq E \leq K$. Each bus's terminal purchase is at most $e_{ij} - r_{ij} = \min(e_{ij}, B) \leq B$, so $H = 2kB$ accommodates the whole interval. Serial charging realizes both common-window budgets; nonnegative subsequent consumption preserves every intermediate SOC constraint. In particular $E = K$ works for every perfect matching, not just a favorable subset.

Give each high A/B service in block i potential $B/2 + 9w_i$ and each low service potential w_i. Write these as row potentials u_i and column potentials v_j; their combined sum is $S_{\max} = kB + 20W$. Every within-block edge satisfies $e_{ij} + 2r_{ij} = u_i + v_j$. An interblock edge has $e_{ij} + 2r_{ij} = B + 3a_i$, with the following slacks:

Origin	Destination	$e_{ij} + 2r_{ij} - u_i - v_j$
High in block i	High in block j	$3B/2 - 9w_j$
High in block i	Low in block j	$2B - w_j$
Low in block i	High in block j	$B/2 + 2w_i - 9w_j$
Low in block i	Low in block j	$B + 2w_i - w_j$

Each slack is at least $\gamma = w_{\max} + 2w_{\min} \geq 3$; the low-to-high bound is limiting and follows from $B/2 \geq 10w_{\max}$. Summing this feasible assignment-potential certificate gives

$$S_M + 2R_M \geq S_{\max},$$

with surplus at least γ if any interblock edge is used. Hence every physical point satisfies $3E + L = S_M + 2E \geq S_{\max}$, and so does every complete-plan mixture.

The original within-block matchings remain feasible with the enlarged terminal budget. The equal mixture of their corners $R = 0$ and $R = 6$ attains $(3, S_{\max} - 9)$, proving $\mathrm{CH} = S_{\max} + 2kf$ by the same affine lower bound as Corollary 36. Within-block physical schedules have minimum excess $9/(2K)$, at exactly those two corners. Every interblock schedule has excess at least $\gamma \geq 3$, strictly greater than $9/(2K) \leq 9/28$. Thus no omitted assignment can improve the physical value; the stated strict gap holds for the full complete-graph opportunity set. The hull optimum remains unique in projected coordinates, and the classical support implication applies.

Finally, for a retained corner R_* use $F_{R_*}(E, L) = 3E + L + (E - R_*)^2/(2K)$. It is coordinatewise strictly increasing and convex as before. Within-block schedules achieve its affine lower bound only at that selected corner; interblock schedules have affine excess at least γ. This proves unique projected optimality of every retained corner despite the complete graph. □

The extension removes infeasible added assignments and graph disconnection as explanations of the obstruction. It does not remove the fleet cap, common-window/charge-only assumptions, scalable connector powers or compulsory edge-task interpretation. Added branches can change efficient portions away from the selected corners; no equality of the full old and new physical frontiers is claimed. The corners remain collinear, the gap shrinks with K, and neither generic hardness, exponential hull complexity, empirical prevalence nor robustness to arbitrary new road options follows. This is an analytically reviewed construction, not a new numerical experiment.

X.1 Strict curvature and local relative robustness

The complete-graph example need not rely on zero curvature in terminal energy or exact saturation of the added edges' post-early capacity. The following strengthening remains an analytical compulsory-task model; it is not an implementation qualification or a perturbation certificate for the public timetable.

Proposition 38 (A strictly convex, locally robust obstruction)**.** *In Proposition 37, keep the within-block edges and put $M = S_{\max}$, $\eta = 1/4$. Replace each interblock task energy by $d_{ij} = B - b_j - \eta$, and increase the terminal power/budget to $H = 2kB + 1$. Choose per-bus acceptance power strictly larger than $\max\{K, H\}$ at the nominal point, for example $\max\{K, H\} + 1$, and hold that acceptance power fixed locally. Here η is an energy margin, not a conversion efficiency. Every perfect matching remains feasible. The coordinatewise strictly increasing, strongly convex supply objective*

$$F(E, L) = 3E + L + \frac{(E - 3)^2}{2K} + \frac{(L - (M - 9))^2}{4M}$$

gives the whole-problem values

$$\mathrm{CH} = M + 2kf, \qquad D - \mathrm{CH} = \frac{9}{2K} + \frac{81}{4M} > 0.$$

For each fixed finite family member, positive gap and feasibility of every perfect matching persist in a sufficiently small relative open neighborhood of its numerical parameters, with topology, timing,

access, fleet cap and charge-only/unit-efficiency law fixed. Economic perturbations are restricted to positive separable quadratics with positive coordinate derivatives at the origin. No neighborhood radius or unchanged optimizer is asserted.

Proof. For an added edge, post-early consumption is $B - \eta < B$ and its early requirement is $r_{ij} = a_i - \eta > 0$. Its early interval $[r_{ij}, a_i]$ has width η. All within-block intervals also have strictly positive widths and strictly feasible post-early consumption under $B \geq 20w_{\max}$. Thus every matching has $R < K$; moreover $S - R = \sum \min(e_{ij}, B) \leq 2kB < H$. Serial common-window charging realizes its entire branch $R \leq E \leq K$, $L = S - E$. Interior allocations can be chosen strictly inside every per-bus early interval, leaving early-budget and preterminal SOC margins as well as terminal-budget slack. Final inventory remains exactly full by the subsequent refill.

The potentials in Proposition 37 remain tight on within-block edges. On an added edge, $e_{ij}+2r_{ij}$ drops by exactly 3η; its potential slack therefore remains at least $3 - 3/4 = 9/4$. Every physical point and every complete-plan mixture obey $3E+L \geq M$, while a physical plan using an interblock edge obeys $3E + L \geq M + 9/4$.

The Hessian of F is $\mathrm{diag}(1/K, 1/(2M))$. Its coordinate derivatives on the nonnegative orthant are bounded below by $3-3/K > 0$ and $1/2+9/(2M) > 0$. The equal mixture of the retained corners $R = 0$ and $R = 6$ gives $(3, M-9)$ and makes both squares zero. Together with the affine bound this proves $\mathrm{CH} = M + 2kf$, uniquely in projected loads. On a within-block branch $L = M - 2R - E$, differentiation gives

$$\frac{dF}{dE} = 2 + \frac{E-3}{K} - \frac{L-(M-9)}{2M} \geq 2 - \frac{3}{K} - \frac{9}{2M} > 0.$$

Its minimum is at $E = R$, with excess $(R-3)^2(1/(2K) + 9/(4M))$. Exactly $R = 0, 6$ minimize it. Their excess is at most $9/28 + 81/160 < 1 < 9/4$, using $K \geq 14$, $M \geq 40$. No interblock plan can improve or tie them, so the stated physical value covers every matching of the complete graph.

For the neighborhood statement, fix this finite labeled model. Allow continuous numerical changes within the admissible nonnegative-energy, nonnegative-fee parameter space, keeping the declared access and serialization conditions. The strict acceptance margin persists for sufficiently small changes of K, H, so fixed acceptance still permits serialization. Zero task energies and $f = 0$ are interpreted in the relative topology. If identities such as $K = \sum_i a_i$ are retained, use the relative topology of that subspace. Strict per-edge interval widths, post-early capacity margins and terminal slack persist simultaneously because only finitely many edges and matchings occur.

For independently varying budgets a matching's branch endpoints are

$$\ell_M = \max\{0, R_M, S_M - H\}, \qquad u_M = \min\{\textstyle\sum_i a_i, K\}.$$

These are continuous, including when $R_M = \sum(e_{ij} - B)_+$ changes its active positive parts, and remain strictly ordered locally. Parameterize each interval by $t \in [0, 1]$. The resulting continuous objectives on a common compact domain have continuous minima; taking the minimum over finitely many matchings proves continuity of D. Their complete-plan hull is generated by the finitely many continuous branch endpoints. Parameterizing it by a fixed simplex proves continuity of CH in the same way, with no nondegeneracy assumption on hull faces. All nearby loads lie in a common bounded box. The nominal strict gap therefore stays positive locally. Positive separable curvature and positive coordinate derivatives at the origin also persist locally, preserving strong convexity and global monotonicity on the nonnegative orthant. □

This continuity argument does not preserve the exact displayed gap formula, its two minimizing corners or its particular hull mixture after perturbation. It gives no radius uniform in k, no robustness to new physics or access options, and no assertion for arbitrary quadratic cross-coefficients. In

particular, it cannot transfer the public binary-energy certificate to a different consumption law: the explicit strict feasibility margins and finite-branch description used here must first be proved for the model in question. Neither practical prevalence, road realization nor computational hardness follows.

Y Implementation qualifications and retained evidence

The analytical claims concern the stated ideal models. Exact rational implementation comparisons use independently formulated physical enumeration and analytical formulas; agreement is an internal cross-check, not external replication. The connector/fee qualification contains twenty fixed cases, and the support-extension qualification fourteen fixed cases, with explicit boundary and corruption controls. These cases qualify implementations; the proofs establish the continuous-domain statements.

The heterogeneous adequate-resource specialization matched independent raw SOC enumeration and threshold formulas on all eight fixed cases; eighteen negative controls and a smoke anchor passed, including tie and continuous- response checks. The fixed single-shared-connector extension separately matched raw SOC enumeration and serial witnesses against sorted-threshold formulas on all fourteen prospectively fixed cases, with twenty-four negative controls and a smoke anchor passing. These include physical collinear-threshold and clipped lower-boundary optima, assignment ties and response continua. Both are internal qualifications, not external replication; the eight-case run concerns the separate n-connector resource policy.

The common-loss extension matched independent raw inventory/sign-polytope enumeration on all ten fixed rational cases, retaining every assignment and infeasible cardinality. Its checks include all projected vertices, exact full-segment attainment and physical replay, rather than a sampled energy grid. The hand-anchor control overlaps one case and is not an additional independent observation. Finite implementation agreement does not prove the continuum theorem; the constructive proofs are separate.

The complete finite-cycle polygons received a distinct eight-case exact qualification. All 34 allowed partial duty matchings and 20 cardinality layers were retained, including resource-infeasible layers. Independently constructed raw bus polygons, Minkowski sums and resource clipping agreed with the five-halfspace formula. All 85 case vertex schedules passed one strict replay of serial actions, SOC and terminal inventories. Both polygon containments were checked, with one fixed matching attaining an entire candidate polygon. The control anchor and all twelve deliberately corrupted alternatives had their expected outcomes; the anchor deliberately overlaps one case and is not an additional independent observation. The cases cover common losses, forbidden edges, zero/tight budgets and one common transition task. Their finite implementation agreement is neither a scalability comparison nor evidence about real-world cycling costs or public-timetable V2G.

Archived outcome and resource records distinguish exact arithmetic from tolerance-qualified native computations. This evidence summary reads stored qualification results without rerunning the calculations. The public positive-gap result uses a source-bound feasible mixture and an independently reconstructed exact 303-node coverage certificate, with 353 classical native proposal calls, followed by a retained-partition refinement with 72 new nodes and 102 new calls. The latter tightly encloses the physical value without supplying a new hull lower. Neither a finite plan bank nor a native status alone establishes the conclusion. Failed strict replays and earlier unsuccessful certification stages remain archived with their costs.

We also packaged the original binary-model evidence with a standalone standard-library Python

checker. One fresh isolated process rechecked both complete physical points, the fixed mixture, the all-fleet exclusions and the combined 375-node proof tree, reproducing the stronger lower bound without a solver. The complete local construction-and-check phase took about 33 seconds. This author-run command qualification is not an independent external replication. Source/data redistribution rights remain under review; the package is currently a local review artifact, not a claimed public release. The source/input identities and trust boundary are in the accompanying certificate index.

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